A Unified Maximum-Likelihood Framework for Signal Reconstruction: From Optimal Filtering to Noise-Aware Linear Autoencoders

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Abstract

Signal reconstruction under structured detector noise is a shared problem of signal processing, statistical latent-variable modeling, and representation learning. This paper develops the linear Gaussian theory of that problem around a single geometric principle: the noise covariance Σ defines the observation-space metric through Σ^{-1}, and statistically efficient reconstruction minimizes the corresponding Mahalanobis (χ^2) objective. Under this geometry, optimal filtering (OF), expectation–maximization PCA (EMPCA), and noise-aware tied linear autoencoders are not separate algorithms but constrained maximum-likelihood projections that differ only in the admissible signal model: a fixed rank-1 template, structured rank-1 extensions, or a learned rank-k subspace. We prove two equivalence results. First, under matched preprocessing, whitening, and weighting, rank-1 EMPCA degenerates exactly to the classical OF amplitude estimator. Second, the Bridge Theorem shows that a tied linear autoencoder trained with the Σ^{-1}-weighted reconstruction loss recovers the EMPCA subspace. An immediate consequence concerns loss design: isotropic mean-squared error is the Gaussian likelihood only for white noise, so under colored or correlated noise MSE can optimize a geometry inconsistent with the assumed noise model. Controlled simulations test the equivalences and expose this metric reversal: under Brownian noise, isotropic PCA attains lower raw MSE yet a worse weighted residual, and the learned subspace separates from the noise-aware one by nearly 90°. The completed evidence bundle adds nine

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multi-seed synthetic validation gates, all of which passed their predeclared criteria, and a public GWOSC stress test on one 256-second GW150914-centered interval. In that interval, the repository Hann-windowed, bias-corrected median PSD matched GWpy exactly at recorded precision, and a shared-FIR follow-up showed implementation identity between explicit FFT convolution and GWpy convolution when both used identical FIR coefficients and normalization. However, held-out real-noise calibration failed under both the global PSD model and the predeclared primary 64-second local PSD model, while the stationary synthetic control for the local-PSD experiment passed. We therefore retain GWOSC event and injection scores as diagnostics only, not as calibrated significance or sensitivity claims. An internal cryogenic-detector calibration sample is used only as a secondary consistency check for the rank-1 equivalence. The resulting linear theory is intended as a likelihood-geometry baseline for comparing scientific reconstruction methods, not as a claim of a universal reconstruction architecture or general real-detector calibration.

1 Introduction

Scientific detectors often record time traces in which a structured response is superposed with electronic, thermal, and environmental fluctuations. The noise is rarely independent and identically distributed: it can be colored in frequency, non-uniform across samples, and correlated across readout channels. The reconstruction task is therefore not merely to smooth a waveform, but to estimate physically meaningful amplitudes, arrival times, or pulse shapes without fitting directions dominated by detector noise. Cryogenic calorimeters provide a concrete example. In superfluid-helium concepts such as DELight, energy can be shared across channels and the measured pulse shape depends on transport, collection, and readout effects. The same statistical issue arises more broadly whenever detector traces have structured signal families and non-isotropic noise.

This reconstruction problem is studied by at least three communities using closely related ideas in different language. In signal processing, matched and optimal filters estimate amplitudes by projecting data onto a known template with inverse-noise weighting. In statistics, principal-component and subspace methods replace the fixed template by a learned low-dimensional signal model, with expectation–maximization PCA (EMPCA) extending PCA to heteroskedastic noise. In machine learning, encoder–decoder models recast reconstruction as representation learning in a latent space. All three address the same task: recover structure from noisy observations by constraining the reconstruction to a low-complexity model.

Despite this close mathematical kinship, optimal filtering (OF), EMPCA, and linear autoencoders (AEs) are usually presented as separate methods. The signal-processing

literature emphasizes estimator optimality for a fixed template under known noise; the statistical literature emphasizes low-rank structure and EM updates; the machine-learning literature emphasizes latent representations and reconstruction losses. What is missing is a single formulation that makes their relationship explicit in the setting most relevant to scientific instrumentation: known or estimated noise covariance, structured waveform families, and the need to compare fixed-template and learned-subspace reconstruction on the same footing.

This paper closes that gap in the linear regime through one organizing claim: *the detector covariance induces the metric of the reconstruction problem.* For Gaussian noise $n \sim \mathcal{N}(0, \Sigma)$, the negative log-likelihood of a candidate reconstruction $\hat{s}$ is the Mahalanobis quadratic form $(x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s})$, so Σ^{-1} defines the observation-space inner product. Under this metric, OF, EMPCA, and the noise-aware tied linear AE are all Σ^{-1}-orthogonal projections; they differ only in the admissible signal subspace — fixed rank-1, structured rank-1 (time-shift and joint-channel variants), or learned rank-k. The key modeling choice is therefore not whether one uses a filter, a subspace method, or an autoencoder, but which constraint is imposed and how the noise geometry is encoded in the loss.

The claim matters because the most common reconstruction loss ignores it. Isotropic mean-squared error (MSE) is the Gaussian maximum-likelihood objective only when $\Sigma = \sigma^2 I$. Under colored or correlated detector noise, MSE is a misspecified likelihood: it rewards fitting high-variance, noise-dominated directions instead of high signal-to-noise directions, and a model can improve its raw MSE while degrading the likelihood-aligned and resolution metrics considered here. The Σ^{-1}-weighted objective aligns the learned representation with the detector's signal-to-noise structure. Loss geometry, in other words, is not a tuning detail.

The scope of this paper is deliberately linear. Restricting to linear subspace models with Gaussian likelihood permits exact statements — closed forms, equivalence theorems, and a controlled ablation in which only the loss geometry varies — and it establishes the likelihood-consistent reconstruction geometry for the stated noise model before nonlinear architectures are introduced. We do not treat nonlinear manifold learning, deep architectures, or variational inference here. Our position is that such models should state their noise likelihood explicitly and be benchmarked against the linear noise-aware baseline.

1.1 Related work

Optimal and matched filtering for detector signal processing. The optimal filter for a known deterministic signal in Gaussian noise is classical: it is the best linear unbiased estimator under the Gauss–Markov conditions and the maximum-likelihood (ML)

estimator under Gaussian noise [1, 2]. Closely related constructions appear as the matched filter [3] and in Wiener filtering [4], although terminology varies with whether the target is a scalar template amplitude or a full random process. For microcalorimeters, the optimal-filter approach was pioneered by Szymkowiak et al. [5], developed for pulse-height extraction by Fixsen et al. [6], and is reviewed by Kurinsky [7] as the standard amplitude estimator for cryogenic sensor readout. We place the fixed-template amplitude estimator explicitly within the Gaussian ML framework and identify it as the rank-1 case of a constrained ML family.

Covariance-aware linear baselines in detector denoising. Recent detector-ML studies continue to compare nonlinear denoisers with Wiener or optimal filtering when the signal model and noise spectrum are available. Parker et al. [8], for example, benchmark supervised, semi-supervised, and unsupervised denoisers for simulated nEXO-type xenon charge measurements against a Wiener-filter baseline. Their unsupervised MC-SURE objective explicitly includes the colored-noise covariance, and their semi-supervised construction models the practical case of approximate signal simulation followed by target-domain noisy measurements. All signal samples in that study are simulated, so it should not be read as a real-data validation of the present framework. Rather, it is consistent with the practical motivation for retaining a covariance-aware linear baseline when assessing more expressive denoisers.

PCA, probabilistic PCA, factor analysis, and EMPCA. Probabilistic PCA [9] and factor analysis [10] perform low-rank signal recovery under Gaussian noise, assuming isotropic and diagonal noise covariance respectively. Bailey [11] introduced EMPCA for PCA under per-observation heteroskedastic weights, and Yu et al. [12] applied a frequency-domain variant to microcalorimeter pulses. Relative to these works, we formulate covariance-weighted low-rank reconstruction as a rank-k constrained Gaussian ML problem, state when it reduces to ordinary PCA after whitening, prove the exact algebraic equivalence between rank-1 weighted PCA and classical OF, and connect the same objective to a noise-aware linear AE.

Linear autoencoders and PCA equivalence. Linear AEs with tied weights recover the PCA subspace under MSE loss [13, 14, 15]. The Bridge Theorem in this paper extends this classical equivalence to the noise-weighted setting: a tied linear AE with Σ^{-1}-weighted reconstruction loss recovers the EMPCA subspace.

Noise-aware and heteroskedastic reconstruction losses. Weighted or noise-dependent losses appear in denoising AEs [16, 17] and heteroskedastic regression [18].

Our contribution is not the Mahalanobis loss itself, but its explicit use as the common comparison geometry linking fixed-template filtering, covariance-weighted low-rank reconstruction, and tied linear autoencoding for detector time series.

Physics-informed and geometric ML positioning. Physics-informed ML typically injects physics through constraint or residual penalties added to a standard objective [19, 20]. Here the physics of the measurement determines the likelihood itself: Σ is not a regularizer but the metric of the problem. This is complementary to geometric deep learning [21], which derives architectures from symmetries of the input domain; here the known noise model instead supplies an observation-space geometry. A compact linear symmetry interpretation is summarized in Section 5.4, with details moved to Appendix E.

1.2 Contributions

1. **Noise-induced metric geometry.** We establish the Σ^{-1}-induced observation-space geometry for linear Gaussian detector reconstruction, under which statistically efficient estimators are Σ^{-1}-orthogonal projections (Section 2).

2. **Unified constrained ML formulation.** We formulate OF, EMPCA, and the noise-aware tied linear AE as constrained minimizers of one Gaussian ML objective, organized as a hierarchy of admissible signal models (Table 1).

3. **OF as the rank-1 case.** We prove that OF is the fixed rank-1 special case and that rank-1 EMPCA degenerates exactly to OF under matched preprocessing, whitening, weighting, and normalization (Theorems 4.1 and 4.2); time-shift and joint-channel OF are structured rank-1 extensions of the same weighted projection principle.

4. **Bridge Theorem.** We prove that a tied linear AE trained with the Σ^{-1}-weighted reconstruction loss recovers the rank-k EMPCA subspace (Theorem 5.1), so the encoder–decoder formulation is the natural endpoint of the linear theory rather than an unrelated architecture.

5. **Loss-design consequence.** We show that isotropic MSE is the Gaussian likelihood only for white noise and can fail under colored or correlated noise, optimizing variance-dominant rather than SNR-relevant directions (Section 6).

6. **Controlled validation and real-data stress testing.** We test the equivalences and the loss-misspecification effect on synthetic data with known covariance, including a metric-reversal experiment in which isotropic PCA wins on raw MSE while losing on the likelihood-aligned metrics reported here. The completed validation

bundle further records nine passed multi-seed synthetic gates, a passed remote reproducibility gate, exact agreement with GWpy for the Hann-windowed median PSD on one GW150914-centered interval, a passed shared-FIR implementation-identity check, and negative held-out GWOSC calibration under both global and predeclared local PSD models. The GWOSC scores are therefore reported as diagnostics rather than calibrated event-significance or injection-sensitivity results. An internal calibration sample is retained only as a secondary consistency check (Section 7).

1.3 Paper organization

Section 2 introduces the unified noise-aware objective, the whitening transform, and the estimator hierarchy. Section 3 develops optimal filtering as the rank-1 constrained case, with time-shift and joint-channel extensions summarized briefly. Section 4 develops EM-PCA as the rank-k constrained case and states the equivalence theorems. Section 5 reframes the same objective in encoder–decoder language and proves the Bridge Theorem. Section 6 draws the loss-design consequence: isotropic MSE assumes white noise, and under colored noise the two objectives can disagree about which reconstruction is better. Section 7 validates the theorems and the loss-misspecification prediction on controlled simulations, with an internal detector calibration sample used as a secondary consistency check, and then reports the transferred evidence-bundle results: passed synthetic gates, passed remote reproducibility, exact GWpy PSD agreement, shared-FIR implementation identity, failed global and local-PSD GWOSC null calibration, and explicit placeholders for future confirmatory real-data claims. Sections 8–10 discuss implications, limitations, and scope. Detailed derivations, full proofs, implementation notes, and additional ablations are collected in Appendices A–H.

2 The Unified Noise-Aware Objective

This section formulates the reconstruction problem as a single Gaussian maximum-likelihood (ML) objective and shows that optimal filtering (OF) and expectation–maximization PCA (EMPCA) are constrained solutions of it. We distinguish three levels of equivalence used throughout the paper: *estimator equivalence* (identical scalar amplitude estimates under matched preprocessing and weighting), *subspace equivalence* (identical signal subspace up to gauge freedom), and *amplitude equivalence* (identical coefficients once projected onto the same subspace with consistent normalization).

Notation and conventions. Vectors are columns. We work in $\mathbb{C}^d$ to cover frequency-domain representations; $\dagger$ denotes the conjugate transpose, and all statements specialize to real-valued data by replacing $\dagger$ with the transpose T. Throughout, $x \in \mathbb{C}^d$ is one observed trace, s its unknown noiseless signal, $\hat{s}$ a candidate reconstruction, s_0 a fixed signal template, $\Sigma \succ 0$ the noise covariance with inverse Σ^{-1} and whitening map $\Sigma^{-1/2}$, $U \in \mathbb{C}^{d\times k}$ a learned rank-k basis with coefficient vector $c \in \mathbb{C}^k$, and $P \in \mathbb{C}^{d\times k}$ the decoder basis of the linear autoencoder. Whitened quantities carry tildes, e.g. $\tilde{x} = \Sigma^{-1/2}x$. Events are indexed by $i = 1, \ldots, N$, frequency bins by k (the rank is also written k; the meaning is always clear from context), and channels by $c = 1, \ldots, C$ where needed.

2.1 Generative model

For each event i we observe $x_i \in \mathbb{C}^d$, a time-domain trace or its frequency-domain coefficients, with the generative model

$$x_i = s_i + n_i, \qquad n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma), \qquad \Sigma \succ 0, \tag{1}$$

where s_i is the unknown noiseless signal for event i and the covariance Σ is known or estimated from dedicated noise data that are independent of the fitted trace. The reconstruction $\hat{s}_i$ is restricted to a specified signal model. In the fixed-template special case, $s_i = a_i s_0$ and $\hat{s}_i = \hat{a}_i s_0$ for a scalar amplitude a_i.

Define the whitening transform

$$\tilde{x}_i = \Sigma^{-1/2}x_i, \qquad \tilde{s} = \Sigma^{-1/2}s_0. \tag{2}$$

In whitened coordinates the noise is isotropic,

$$\tilde{x}_i = \Sigma^{-1/2}s_i + \tilde{n}_i, \qquad \tilde{n}_i \sim \mathcal{N}_{\mathbb{C}}(0, I), \tag{3}$$

so every Σ^{-1}-weighted projection becomes an ordinary orthogonal projection. Stacking events row-wise gives the whitened data matrix $\tilde{X} \in \mathbb{C}^{N\times d}$ with rows $\tilde{x}_i^\dagger$.

2.2 The Gaussian ML objective

For a single trace x under model (1), the Gaussian negative log-likelihood is, up to additive constants,

$$\chi^2(\hat{s}) = (x - \hat{s})^\dagger \Sigma^{-1}(x - \hat{s}). \tag{4}$$

Every estimator in this paper minimizes objective (4); the methods differ only in the constraint placed on the admissible reconstruction $\hat{s}$.

Optimal filtering: fixed rank-1 constraint. For a known template s_0 and unknown scalar amplitude A, the constraint is $\hat{s} = As_0$. Minimizing (4) over A gives the generalized least-squares estimator

$$\hat{A}_{\mathrm{OF}} = \frac{s_0^\dagger \Sigma^{-1} x}{s_0^\dagger \Sigma^{-1} s_0}, \tag{5}$$

which is exactly the optimal filter in covariance form. For stationary noise in Fourier space, Σ is diagonal and (5) reduces to the familiar PSD-weighted OF formula (Section 3).

EMPCA: learned rank-k constraint. For a learned basis $U = [u_1, \ldots, u_k] \in \mathbb{C}^{d\times k}$, the constraint is $\hat{s} = Uc$, and

$$\chi^2(c) = (x - Uc)^\dagger \Sigma^{-1} (x - Uc) \tag{6}$$

is minimized by the weighted normal equations

$$\hat{c} = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x, \qquad \hat{s}_{\mathrm{EMPCA}} = U\hat{c}. \tag{7}$$

Equation (7) makes the connection to OF explicit: OF is the special case $k = 1$ with $U = s_0$ fixed.

2.3 Whitening as the geometric normal form

Whitening converts the Σ^{-1}-weighted inner product into the standard Euclidean one: for any $u, v \in \mathbb{C}^d$,

$$u^\dagger \Sigma^{-1} v = (\Sigma^{-1/2} u)^\dagger (\Sigma^{-1/2} v) = \tilde{u}^\dagger \tilde{v}. \tag{8}$$

After whitening, the noise covariance is the identity and all estimators in this paper become ordinary orthogonal projections; this is the simplification that makes the geometric statements of Sections 3–5 exact.

In the most common detector setting the noise is wide-sense stationary, so Σ is circulant and diagonalized by the DFT:

$$\Sigma = F^\dagger \mathrm{diag}(J_1, \ldots, J_d) F, \tag{9}$$

where $J_k = \mathbb{E}[|N_k|^2]$ is the noise power spectral density (PSD) at frequency bin k. Whitening is then element-wise division in the frequency domain,

$$\tilde{X}_k = X_k / \sqrt{J_k}. \tag{10}$$

For non-circulant Σ (short traces, non-stationary noise, correlated channels), $\Sigma^{-1/2}$ is computed by eigendecomposition or Cholesky factorization; numerically small PSD bins require a regularization floor or frequency masking. These implementation details, which do not affect the theory, are collected in Appendix C.

2.4 The estimator hierarchy

Table 1 summarizes the estimator family studied in this paper. Each row is a constrained minimizer of the same Gaussian ML objective (4); the rows differ only in how the admissible reconstruction is constrained.

Table 1: Hierarchy of constrained ML estimators under the unified noise-aware objective. All methods minimize $(x - \hat{s})^\dagger \Sigma^{-1} (x - \hat{s})$. "Degeneracy condition" states when a row coincides with the basic OF row.

Estimator	Constraint on $\hat{s}$	Subspace dim.	Solution form	Degeneracy condition
Basic OF	$\hat{s} = As_0,\ s_0$ fixed	$k = 1$ (fixed)	$\hat{A} = (s_0^\dagger \Sigma^{-1} x)/(s_0^\dagger \Sigma^{-1} s_0)$	Signal is exactly $\propto s_0$
Time-shift OF	$\hat{s} = Ae^{-i\omega t_0} s_0,\ t_0$ unknown	$k = 1$ (shift manifold)	$\hat{t}_0 = \arg\max_{t_0} \lvert\hat{A}(t_0)\rvert$	No timing jitter; collapses to basic OF
Joint-channel OF	$\hat{s} = As,\ s \in \mathbb{C}^{Cd}$ joint template	$k = 1$ (joint)	$\hat{A} = (s^\dagger \Sigma^{-1} x)/(s^\dagger \Sigma^{-1} s)$	Channels uncorrelated; factorizes per channel
Rank-k EMPCA	$\hat{s} = Uc,\ U \in \mathbb{C}^{d \times k}$ learned	k (learned)	$\hat{c} = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x$	$k = 1$, span(U) = span(s_0): equals basic OF
Noise-aware linear AE	$\hat{s} = PP^\dagger \Sigma^{-1} x,\ P^\dagger \Sigma^{-1} P = I$	k (learned)	P^* = top-k eigenvectors of $\hat{R}_x \Sigma^{-1}$	Equivalent to EMPCA at convergence (Theorem 5.1)

Table 2 states the same comparison from an evaluation perspective. Ordinary PCA is included because it is the Euclidean baseline used in the loss-geometry experiments, even though it does not minimize the covariance-weighted objective unless the data have first been correctly whitened.

Table 2: Methods compared under a shared reconstruction task. "Uses covariance" refers to the fitted objective, not merely to post-hoc evaluation.

Method	**Model and output**	**Uses Σ**	**Known template**	**Colored or correlated noise**	**Likelihood interpretation**
OF / matched filter	Fixed rank-1; amplitude and template reconstruction	Yes	Yes	Yes, with valid Σ or PSD	Gaussian ML for the template amplitude
Ordinary PCA	Learned rank-k Euclidean subspace and scores	No	No	Only after exact whitening	Gaussian ML only for $\Sigma = \sigma^2 I$
Weighted PCA / EMPCA	Learned rank-k weighted subspace and coefficients	Yes	No	Yes	Gaussian ML for the specified covariance or weights
Noise-aware tied linear AE	Rank-k subspace, latent code, and reconstruction	Yes	No	Yes, in this paper	Same linear-Gaussian objective as weighted PCA / EMPCA

Three structural observations follow. First, constraint complexity increases monoton-

ically down the table: basic OF is the most constrained (one free parameter, fixed geometry) and the noise-aware linear AE the least (a full learned k-dimensional subspace). Second, the noise model enters identically in every row through the same quadratic form — the methods differ in their signal model, not in how they treat noise. Third, degeneracy propagates upward: each row collapses to the row above when its extra freedom is unused. Sections 3–5 prove these degeneracies; exact improvement over OF occurs only when the true signal family has dimensions a fixed template cannot capture.

2.5 Multi-channel extension

Detectors with C simultaneously read-out channels are covered by stacking the channel traces into a joint vector $x \in \mathbb{C}^{Cd}$ with block-structured covariance $\Sigma_{\text{joint}} \in \mathbb{C}^{Cd \times Cd}$, whose diagonal blocks are intra-channel covariances and whose off-diagonal blocks carry cross-channel noise correlations. The Gaussian ML objective (4) is unchanged in form; the rank-1 constraint gives joint-channel OF (Section 3.3) and the rank-k constraint gives multi-channel EMPCA. When cross-channel correlations vanish and each channel covariance is diagonal, the joint objective decouples into C independent single-channel problems. The full block-covariance construction and whitening details are given in Appendix B.2.

3 Optimal Filtering as Rank-1 Constrained ML

3.1 The rank-1 constraint and OF estimator

Specializing the unified objective (4) to the fixed-template constraint $\hat{s} = As_0$ in the stationary single-channel case, where Σ is diagonal in frequency space with PSD entries J_k, gives

$$\chi^2(A) = \sum_k \frac{|X_k - AS_k|^2}{J_k}, \tag{11}$$

the discrete Gaussian log-likelihood for uncorrelated frequency bins [1]; this frequency-domain form for microcalorimeters was first stated by Szymkowiak et al. [5]. The stationarity condition $\partial\chi^2/\partial A^* = 0$ yields

$$\hat{A} = \frac{\sum_k \frac{S_k^* X_k}{J_k}}{\sum_k \frac{|S_k|^2}{J_k}}, \tag{12}$$

which is the covariance-form estimator (5) in frequency coordinates. Defining the filter kernel and normalization

$$\Phi_k = \frac{S_k^*}{J_k}, \qquad N_\Phi = \sum_k \frac{|S_k|^2}{J_k}, \qquad \Phi_k^{(0)} = \frac{\Phi_k}{N_\Phi}, \tag{13}$$

the estimate $\hat{A} = \sum_k \Phi_k^{(0)} X_k$ is the filtered inner product of the data with the normalized kernel — the discrete matched filter [3]. Geometrically, $\hat{s}_{\mathrm{OF}} = \hat{A} s_0$ is the Σ^{-1}-orthogonal projection of x onto the fixed one-dimensional template subspace.

Assumption boundary. The covariance-form estimator requires a known template direction and a known or independently estimated positive-definite noise covariance. Stationarity is needed only for the convenient diagonal PSD representation in Eq. (11); the generalized least-squares expression remains valid for non-stationary or cross-channel covariance when the full matrix is used. Gaussianity is required for the likelihood and efficiency claims, while the best-linear-unbiased statement needs only the corresponding second-moment assumptions. OF is therefore optimal for this fixed-template linear problem. It is not a solution to general nonlinear waveform reconstruction, unknown signal morphology, or covariance mismatch.

3.2 Estimator variance and energy resolution

The OF estimator is unbiased, and its variance

$$\mathrm{Var}(\hat{A}) = \frac{1}{N_\Phi}, \qquad \sigma_A = \frac{1}{\sqrt{N_\Phi}}, \tag{14}$$

attains the Cramér–Rao bound (CRB) for the Gaussian model, making OF the minimum-variance unbiased estimator for a fixed template; the full Fisher information calculation is given in Appendix A. When the amplitude calibrates to deposited energy, $A = E/E_0$, the baseline energy resolution is

$$\sigma_E = \frac{E_0}{\sqrt{N_\Phi}} = E_0 \left(\sum_k \frac{|S_k|^2}{J_k} \right)^{-1/2}. \tag{15}$$

Equation (15) carries the design intuition: resolution improves where signal power is large and noise PSD small, and N_Φ is simultaneously the filter normalization and the total signal-to-noise budget available to any rank-1 estimator. Empirical verification of the CRB and of the $\sigma_E \propto \sqrt{\text{noise power}}$ scaling is reported in Appendix F.

3.3 Extensions: the same projection principle

Two standard OF extensions are structured applications of the same rank-1 weighted projection principle, and both are absorbed naturally by the hierarchy of Table 1.

Time-shift OF. For an unknown arrival time, the admissible signals form the one-parameter manifold $\mathcal{M} = \{e^{-i\omega t_0} s_0 : t_0 \in \mathbb{R}\}$. Profiling the amplitude at each candidate shift gives $\hat{A}(t_0) = \sum_k e^{i\omega_k t_0} \Phi_k^{(0)} X_k$ — the inverse DFT of the filtered spectrum — and the ML delay maximizes the filtered amplitude, $\hat{t}_0 = \arg\max_{t_0} |\hat{A}(t_0)|$ [7]. For small shifts, a Taylor expansion shows $\mathcal{M}$ is approximately contained in $\mathrm{span}\{s_0, \partial_t s_0, \partial_t^2 s_0, \dots\}$, so a low-rank learned subspace can absorb timing jitter; this is exactly the mechanism by which rank-2 EMPCA removes timing bias in Section 7. The full profiling derivation and the formal degeneracy statement (Proposition B.1) are given in Appendix B.1.

Joint-channel correlated OF. For C channels stacked into $x, s \in \mathbb{C}^{Cd}$ with full joint covariance Σ, the GLS estimator $\hat{A}_{\mathrm{JOF}} = (s^\dagger \Sigma^{-1} x)/(s^\dagger \Sigma^{-1} s)$ is algebraically the rank-1 weighted subspace estimator (7) with $U = s$. Cross-channel noise correlations are handled identically in OF and EMPCA, through the same quadratic form; the formal statement (Proposition B.2) and the block-covariance details are in Appendix B.2.

3.4 Limitation: the fixed-template assumption

The OF derivation assumes every signal is a scalar multiple of the *same* known template. Realistic pulse families violate this in two generic ways: timing jitter spreads the signal over the shift manifold, and physical effects (position-dependent response, geometry inhomogeneity, channel cross-talk, saturation) make the pulse shape itself vary event-to-event. When the template direction s_0 is misaligned with the true signal s, the OF estimate is biased,

$$\mathbb{E}[\hat{A}_{\mathrm{OF}}] = A \cdot \frac{s_0^\dagger \Sigma^{-1} s}{s_0^\dagger \Sigma^{-1} s_0} \neq A \quad (\text{if } s_0 \not\propto s), \tag{16}$$

with effective SNR loss $\cos^2 \theta_w$, the squared cosine of the whitened-space angle between s_0 and s. A template-mismatch experiment quantifying this bias is reported in Appendix F. Replacing the fixed template by a *learned* k-dimensional subspace removes this assumption: when the true signal family lies in the learned subspace, the estimator is unbiased and attains the CRB of the subspace-constrained problem. This motivates the rank-k extension developed next.

4 EMPCA as Rank-k Constrained ML

4.1 The rank-k constraint

The optimal filter assumes a one-dimensional signal family. Realistic families have more structure: pulse shapes vary with interaction position, timing jitter generates a continuum of shifted templates, and multi-mode excitations populate several directions simultaneously. The natural relaxation constrains the signal to a *learned* k-dimensional subspace. For a single observation $x \in \mathbb{C}^d$ the rank-k constrained ML problem is

$$\min_{U,\,c} \ \chi^2(c;\,U) = (x - Uc)^\dagger \Sigma^{-1}(x - Uc), \qquad \text{subject to} \quad U^\dagger \Sigma^{-1} U = I_k, \tag{17}$$

and over N events with coefficient matrix $C = [c_1, \ldots, c_N] \in \mathbb{C}^{k \times N}$,

$$\min_{U,\,C} \ \mathcal{L}(U, C) = \frac{1}{N} \sum_{i=1}^{N} (x_i - Uc_i)^\dagger \Sigma^{-1}(x_i - Uc_i), \qquad U^\dagger \Sigma^{-1} U = I_k. \tag{18}$$

In whitened coordinates ($\tilde{x}_i = \Sigma^{-1/2} x_i$, $\tilde{U} = \Sigma^{-1/2} U$, $\tilde{U}^\dagger \tilde{U} = I_k$) the objective reduces to a standard low-rank approximation problem,

$$\mathcal{L}(U, C) = \frac{1}{N} \sum_{i=1}^{N} \|\tilde{x}_i - \tilde{U} c_i\|_2^2, \tag{19}$$

whose minimizer over subspaces is the span of the top-k eigenvectors of the whitened sample covariance $\Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2}$, with $\hat{R}_x = N^{-1} \sum_i x_i x_i^\dagger$. Standard PCA is the ML estimator of this subspace only when $\Sigma = \sigma^2 I$; EMPCA restores ML optimality for general Σ by carrying the weight Σ^{-1} through the estimation.

Algorithm. EMPCA solves (18) by alternating minimization [11]: the E-step solves the weighted normal equations for each c_i with U fixed (closed form, Eq. (7)), and the M-step solves a weighted regression for U with C fixed, followed by re-orthonormalization in the Σ^{-1} inner product. The objective (18) is biconvex, each step is an exact block minimization, and the iteration decreases the objective monotonically and converges to the stationary set, which for this problem coincides with the top-k eigenspace up to gauge (Theorem C.1, Appendix C.5). Implementation details — the frequency-domain variant of Yu et al. [12], initialization, model-order selection, and conditioning safeguards — are collected in Appendix C.

Ordinary PCA versus weighted PCA / EMPCA. Ordinary PCA minimizes Euclidean reconstruction error in the original coordinates. The weighted problem above

minimizes that error after the noise geometry has been applied. When every event shares the same positive-definite covariance Σ, exact whitening reduces the weighted problem to ordinary PCA on $\tilde{X} = X\Sigma^{-1/2}$; this is an equivalence between weighted PCA in detector coordinates and ordinary PCA in whitened coordinates. It is not, in general, an equivalence to ordinary PCA on the unwhitened traces. The latter coincides with the weighted solution when $\Sigma = \sigma^2 I$, or accidentally when the weighting leaves the relevant eigenspace and its ordering unchanged. With event-dependent variances, missing samples, or weights Σ_i^{-1}, no single global whitening need exist and the alternating EMPCA updates are required. Consequently, PCA and EMPCA should agree in the white-noise control but should not be expected to agree under genuinely colored weighting. Numerically identical outputs in that regime would require an explicit audit of whether the weights entered the fit, whether both methods received already whitened data, and whether a degenerate eigenspace obscured the comparison.

Gauge freedom. The basis U is identified only up to a unitary rotation $U \mapsto UQ$, $Q \in \mathcal{U}(k)$: the reconstruction $\hat{s} = Uc$ and the subspace are gauge-invariant, the individual coefficients are not. Physical amplitude estimates therefore use gauge-invariant functionals, either the whitened reconstruction norm $\|\hat{s}\|_{\Sigma^{-1}}$ or the projection onto a calibration template,

$$\hat{A} = \frac{s_0^\dagger \Sigma^{-1} \hat{s}}{s_0^\dagger \Sigma^{-1} s_0}, \tag{20}$$

which reduces to the OF estimator when $k = 1$ and $\mathrm{span}(U) = \mathrm{span}(s_0)$. The geometric structure behind this gauge freedom (the Grassmannian picture of subspace optimization) is developed in Appendix E.

4.2 Equivalence theorems: rank-1 EMPCA and basic OF

We can now state the first main result. Recall the three levels of equivalence from Section 2: subspace equivalence implies amplitude equivalence implies estimator equivalence, provided weighting and normalization conventions are matched.

Theorem 4.1 (Rank-1 OF–EMPCA equivalence, population level)**.** *Assume (i) the generative model $x_i = a_i s + n_i$ with fixed template s; (ii) Gaussian noise $n_i \sim \mathcal{N}(0, \Sigma)$ with known covariance; (iii) identical whitening, baseline subtraction, and preprocessing conventions; and (iv) a one-dimensional true signal subspace. Then, at the population level (sample covariance equal to its expectation):*

1. *the rank-1 EMPCA subspace equals* $\mathrm{span}(\tilde{s})$ *in whitened space;*

2. *the EMPCA coefficient estimator equals the generalized least-squares / optimal-filter estimator;*

3. *hence rank-1 EMPCA and OF are estimator-equivalent.*

Proof sketch. In whitened space the population covariance is the spiked identity $\mathbb{E}[\tilde{x}\tilde{x}^\dagger] = \mathbb{E}[a^2]\,\tilde{s}\tilde{s}^\dagger + I$, whose leading eigenvector is $\tilde{s}/\|\tilde{s}\|$. Rank-1 EMPCA recovers this eigenvector, and the rescaled projection coefficient is $\hat{a}_i = (s^\dagger\Sigma^{-1}x_i)/(s^\dagger\Sigma^{-1}s)$, which is exactly (5). The full proof, including the eigenvalue gap and the normalization bookkeeping, is given in Appendix D.1. □

Theorem 4.1 is a population-level statement; finite-sample deviations are controlled by the Davis–Kahan bound discussed in Section 4.4. The next result is exact at any sample size, for a fixed basis.

Theorem 4.2 (Exact degeneracy to basic OF)**.** *Assume $k = 1$; the EMPCA basis spans the template direction, $u = \gamma s_0$ with $\gamma \neq 0$; OF and EMPCA use the same weighting Σ^{-1}; and preprocessing and alignment conventions are matched. Then rank-1 EMPCA and basic OF yield identical reconstructed signals $\hat{s}$ and identical weighted residuals.*

Proof sketch. Substituting $u = \gamma s_0$ into the rank-1 coefficient $\hat{c} = (u^\dagger\Sigma^{-1}x)/(u^\dagger\Sigma^{-1}u)$ gives $\hat{c} = \hat{A}_{\mathrm{OF}}/\gamma$, hence $\hat{s}_{\mathrm{EMPCA}} = \hat{c}\,u = \hat{A}_{\mathrm{OF}}s_0 = \hat{s}_{\mathrm{OF}}$. The scalar γ is the rank-1 gauge freedom: it changes the coefficient convention but not the reconstruction. Details in Appendix D.1. □

4.3 Why $k > 1$ helps: the χ^2 improvement

In whitened coordinates, OF projects onto a one-dimensional subspace $\mathcal{S}_1$ and rank-k EMPCA onto a k-dimensional subspace $\mathcal{U}_k$. If $\mathcal{S}_1 \subseteq \mathcal{U}_k$, projection theory gives

$$\chi^2_{\mathrm{EMPCA},k} \leq \chi^2_{\mathrm{OF}} : \tag{21}$$

a larger admissible subspace can only reduce the weighted residual. The physical content is that pulse-shape variation, timing jitter, and channel-coupling modes are additional signal directions that a fixed template conflates into a single amplitude.

Caveat: containment is not automatic. The bound (21) requires the template direction to lie in the learned subspace. Unsupervised EMPCA includes $\mathcal{S}_1$ whenever the signal is the dominant variance direction in whitened space (specifically, when $\mathbb{E}[a^2]\|\tilde{s}\|^2$ exceeds the $(k{+}1)$-st noise eigenvalue); below that SNR, or with unfavorable initialization, containment can fail in finite samples. Template-seeded initialization or an explicit containment constraint restores the guarantee. Section 7 verifies empirically that containment holds in the benchmark regime.

4.4 Geometric interpretation and finite-sample stability

After whitening, both estimators are orthogonal projectors:

$$\hat{\tilde{x}}_{\mathrm{OF}} = P_{\tilde{s}_0}\tilde{x}, \qquad \hat{\tilde{x}}_{\mathrm{EMPCA}} = P_{\tilde{U}}\tilde{x}, \qquad P_{\tilde{U}} = \tilde{U}(\tilde{U}^\dagger\tilde{U})^{-1}\tilde{U}^\dagger. \tag{22}$$

OF projects onto a fixed rank-1 direction; EMPCA onto a learned rank-k subspace; when $k = 1$ and the learned direction equals the template direction, the projectors coincide. Because the objective depends on $\tilde{U}$ only through the projector, EMPCA optimizes over the Grassmannian of k-dimensional subspaces; that geometric picture, and its connection to the latent gauge group, is developed in Appendix E.

In finite samples, the estimated subspace deviates from the population one by an angle controlled by the Davis–Kahan theorem [22]:

$$\sin\theta \le \frac{\|\hat{R}_x - R_x\|_{\mathrm{op}}}{\Delta_\lambda}, \tag{23}$$

where Δ_λ is the spectral gap between signal and noise eigenvalues of the whitened covariance. For Gaussian noise the numerator is $O(\sqrt{d/N})$ with high probability, and for the rank-1 model the gap grows with SNR as $\Delta_\lambda \approx N_\Phi\,\mathbb{E}[a^2]$. This explains the near-unity weighted cosines observed on the internal calibration sample in Section 7: at high SNR and large N, the sample subspace is extremely close to the population subspace. The extended stability discussion is given in Appendix E.

From algorithm to objective. As defined above, EMPCA is an algorithmic procedure — alternating E- and M-steps. For comparing estimators at the level of the loss, and for replacing the linear encoder by a nonlinear one while preserving the noise-aware structure, an objective-first formulation is preferable. The next section provides it: the same estimator is the minimizer of a noise-aware tied linear autoencoder loss.

5 The Noise-Aware Linear Autoencoder and the Bridge Theorem

5.1 The encoder–decoder language

We now translate the weighted-projection estimators of Sections 3 and 4 into encoder–decoder language. A linear autoencoder consists of an encoder $E : \mathbb{C}^d \to \mathbb{C}^k$ producing a latent code $z = Ex$ and a decoder $D : \mathbb{C}^k \to \mathbb{C}^d$ producing the reconstruction $\hat{x} = Dz$. This is the division of labor already present in OF and EMPCA: OF computes one weighted coefficient along a fixed template and decodes it into a rank-1 waveform;

EMPCA computes k weighted coefficients in a learned basis. The autoencoder formulation introduces no new estimator by itself — the question is which loss and which encoder–decoder relationship recover the Gaussian ML projection.

For ordinary Euclidean linear AEs, "tied weights" means the encoder is the Hermitian transpose of the decoder. In detector coordinates this is the wrong adjoint, because the data-space inner product is set by Σ^{-1}, not the Euclidean metric. The noise-aware tied form is

$$D(z) = Pz, \qquad E(x) = P^\dagger \Sigma^{-1} x, \qquad P^\dagger \Sigma^{-1} P = I_k, \tag{24}$$

i.e. the encoder is the Σ^{-1}-adjoint of the decoder. This section shows that this architecture, trained with the Mahalanobis reconstruction loss, is exactly EMPCA.

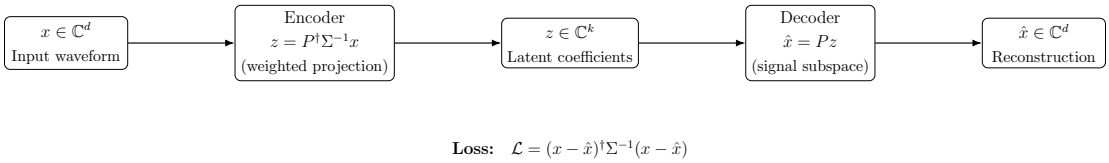


Figure 1: The noise-aware tied linear autoencoder. The encoder is the Σ^{-1}-adjoint of the decoder, and the training loss is the Gaussian χ^2 objective (4).

5.2 The noise-aware loss and its minimizer

With the tied form (24), the reconstruction is $\hat{x}_i = PP^\dagger \Sigma^{-1} x_i$ and the noise-aware reconstruction loss over N events is

$$\mathcal{L}(P) = \frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{x}_i)^\dagger \, \Sigma^{-1} \, (x_i - \hat{x}_i). \tag{25}$$

Expanding with $P^\dagger \Sigma^{-1} P = I_k$,

$$\mathcal{L}(P) = \frac{1}{N} \sum_{i=1}^{N} x_i^\dagger \Sigma^{-1} x_i \; - \; \mathrm{Tr}\big(P^\dagger \Sigma^{-1} \hat{R}_x \Sigma^{-1} P\big), \qquad \hat{R}_x = \frac{1}{N} \sum_i x_i x_i^\dagger, \tag{26}$$

so minimizing $\mathcal{L}(P)$ is a Rayleigh–Ritz problem whose optimum spans the top-k eigenvectors of the Σ^{-1}-weighted covariance operator — precisely the EMPCA subspace. Equivalently, in whitened coordinates ($W = \Sigma^{-1/2} P$, $W^\dagger W = I_k$) the loss is the classical tied linear AE loss $\|\tilde{X} - \tilde{X} W W^\dagger\|_F^2$, minimized by the top-$k$ right singular vectors of $\tilde{X}$.

5.3 The Bridge Theorem

Theorem 5.1 (Bridge Theorem: noise-aware linear AE $\equiv$ EMPCA)**.** *Let $x_i = s_i + n_i \in \mathbb{C}^d$ for $i = 1, \ldots, N$, where $n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$ with $\Sigma \succ 0$ known. Consider the noise-aware tied*

linear autoencoder loss

$$\mathcal{L}(P) = \frac{1}{N}\sum_{i=1}^{N}(x_i - PP^\dagger\Sigma^{-1}x_i)^\dagger\,\Sigma^{-1}\,(x_i - PP^\dagger\Sigma^{-1}x_i), \qquad P^\dagger\Sigma^{-1}P = I_k. \tag{27}$$

Then:

1. *Every minimizer P^* of $\mathcal{L}(P)$ satisfies* $\mathrm{range}(P^*) = \mathrm{range}(\hat{U})$*, where $\hat{U}$ is the rank-k EMPCA solution with weight Σ^{-1}, up to a unitary rotation within the k-dimensional subspace.*

2. *Equivalently, P^* spans the top-k eigenvectors of the generalized eigenproblem $\hat{R}_x\Sigma^{-1}v = \lambda v$.*

3. *The minimum value is $\mathcal{L}^* = \mathrm{Tr}(\Sigma^{-1}\hat{R}_x) - \sum_{j=1}^{k}\lambda_j$, with $\lambda_1 \geq \cdots \geq \lambda_k$ the top eigenvalues of $\Sigma^{-1}\hat{R}_x$.*

Proof sketch. Whitening maps the constraint to $W^\dagger W = I_k$ and the loss to $\frac{1}{N}\|\tilde{X}(I - WW^\dagger)\|_F^2$. By Rayleigh–Ritz the minimizer is the top-k eigenspace of $\Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, which maps back to the generalized eigenspace of $\hat{R}_x\Sigma^{-1}$; EMPCA converges to the same eigenspace (Section 4). The full proof, including the minimum value and uniqueness up to gauge, is given in Appendix D.3. □

Remark 5.1. *The Bridge Theorem is a finite-sample statement: it holds for any $N \geq 1$, not only in the population limit. The minimizer is unique only up to the latent gauge rotation $P^* \mapsto P^*Q$, $Q \in \mathcal{U}(k)$; physical quantities depend only on the reconstruction $P^*P^{*\dagger}\Sigma^{-1}x$ and are gauge-invariant. Experiment B (Section 7) verifies the theorem numerically: the minimum principal-angle cosine between the EMPCA subspace and the exact AE minimizer exceeds* 0.9999 *for all ranks tested.*

The Bridge Theorem closes the equivalence chain of the paper. The noise-aware tied linear AE is not an unrelated neural architecture: in the linear tied case it is the encoder–decoder *parameterization* of the same weighted low-rank reconstruction problem solved by EMPCA, which in turn degenerates to OF at rank 1. It is therefore the correct linear baseline against which nonlinear autoencoders should be compared.

5.4 Why this architecture: a symmetry summary

There is a structural reason the tied Σ^{-1}-adjoint architecture and the χ^2 loss appear together. Two symmetry groups act on the problem: the *noise isometry group* of linear maps preserving the Σ^{-1} inner product, and the *latent gauge group* $\mathcal{U}(k)$ rotating the basis within the learned subspace. Requiring the quadratic reconstruction loss to be

invariant under all noise isometries forces the loss weight to be proportional to Σ^{-1} (the Mahalanobis χ^2 loss, which is also the Fisher information metric of the Gaussian family [23]); requiring the reconstruction to be the metric-orthogonal projection onto the learned subspace forces the encoder to be the Σ^{-1}-adjoint of the decoder. Tied weights are thus not a regularization heuristic but an adjointness condition; untied linear encoders generally define oblique projections and break the equivalence to EMPCA. The formal statement (Theorem E.1) and its proof, together with the Grassmannian geometry of the subspace family, are given in Appendix E.

5.5 Scope of the equivalence claim

The equivalences of this section are rigorous for linear subspace models, Gaussian noise likelihood, matched whitening/weighting and preprocessing conventions, and identical rank k. They do not automatically extend to nonlinear autoencoders or mismatched objectives; what does transfer is the loss geometry, which is the subject of the next section.

6 Loss Geometry: Why MSE Can Be Wrong

The Bridge Theorem has an immediate consequence for machine-learning loss design. Under the measurement model $x_i = s_i + n_i$, $n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$, the Gaussian negative log-likelihood of a reconstruction $\hat{x}_i$ is, up to constants,

$$\mathcal{L}_{\Sigma^{-1}} = \frac{1}{N} \sum_{i=1}^{N} (x_i - \hat{x}_i)^\dagger \Sigma^{-1} (x_i - \hat{x}_i), \tag{28}$$

whereas the standard isotropic loss used by ordinary PCA and most autoencoders is

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_{i=1}^{N} \|x_i - \hat{x}_i\|_2^2. \tag{29}$$

The two coincide (up to scale) exactly when $\Sigma = \sigma^2 I$. Isotropic MSE is therefore not a neutral default: it is the Gaussian likelihood for white noise, and under colored, heteroskedastic, or correlated detector noise it is a *misspecified* likelihood.

The geometric mechanism. The effect of the misspecification is not a rescaling of the objective; the two losses define different inner products and hence different notions of which directions in waveform space matter. Isotropic MSE weights every time or frequency bin equally, so the learned subspace is driven by raw waveform variance — which, under

colored noise, is typically dominated by noise-rich, low-information directions. The Σ^{-1}-weighted loss weights residuals by inverse noise power, so the learned subspace is driven by signal-to-noise structure: noisy directions are down-weighted and informative directions emphasized. The same model class, rank, data, and optimizer can therefore converge to substantially different subspaces under the two losses.

Metric reversal. A concrete failure mode follows. Because each objective is optimal in its own geometry, a method can simultaneously achieve *lower raw MSE* and a *worse weighted residual* (equivalently, worse held-out Gaussian likelihood and worse detector-relevant resolution), while learning a subspace nearly orthogonal — in the whitened metric — to the one required for efficient estimation. We refer to this as *metric reversal*: the ranking of methods reverses depending on whether the evaluation metric respects the noise geometry. Selecting models by raw MSE under colored noise can therefore systematically prefer a reconstruction misaligned with the assumed Gaussian noise model.

Testable predictions. The argument yields two predictions that the experiments of Section 7 test directly, keeping the model class, latent rank, training data, preprocessing, and train/test split fixed while changing only the loss geometry:

- under white noise, isotropic PCA and noise-aware EMPCA should recover the same subspace and agree on all metrics;
- under colored noise, the objectives should diverge: the Σ^{-1}-weighted objective should win on weighted residual, held-out likelihood, and energy resolution, while isotropic PCA may win on raw MSE — the metric it optimizes.

The white-noise case acts as an internal control: if the two objectives coincide when $\Sigma \propto I$ but separate under colored noise, the observed effect is attributable to loss geometry rather than to rank, optimization, or preprocessing differences.

7 Experimental Validation

The experiments are organized around the paper's claims rather than around an exhaustive diagnostic sweep. The controlled synthetic benchmark, for which the covariance and signal rank are known, is the foundation of the validation. Experiments B–E test the equivalence and loss-geometry claims within that controlled setting. Experiment A uses an internal cryogenic-detector calibration sample only as a secondary consistency check for the rank-1 OF–EMPCA relation; no central claim depends on access to that sample. Each experiment is reported as question, setup, metric, result, and interpretation. Supporting

diagnostics — CRB and resolution-scaling checks, template-mismatch and time-shift studies, residual calibration versus rank, convergence traces, bootstrap confidence intervals, and the detector noise-budget study — are collected in Appendix F.

This revision also reports the completed results from the transferred evidence bundle. Those records are treated as the evidence source for the public-data claims below: the S1–S9 synthetic gate sweeps, Stage 0 remote reproducibility, GWOSC data coverage and checksums, the GWpy PSD reference comparison, the shared-FIR implementation-identity follow-up, and the global/local-PSD GWOSC held-out null calibration tests. Results that are pending, absent, diagnostic-only, or not solidly passed are marked by explicit placeholders rather than converted into claims.

7.1 Common benchmark and metrics

Benchmark domain. Simulated traces follow the generative model $x_i = a_i s + n_i$, $n_i \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma)$, with a decaying-exponential pulse template ($\tau = 3\,\text{ms}$, 32 768 samples at 10 kHz) and three stationary noise models: pink ($J(f) \propto 1/f$), white, and Brownian ($J(f) \propto 1/f^2$), spanning the colored-noise regimes relevant to cryogenic microcalorimeters [5]. A controlled multi-dimensional family adds mild timing jitter so that a second principal component (the pulse derivative) carries visible variance. A separate internal K-α calibration sample from a cryogenic microcalorimeter is used only in Experiment A and in the numerical parameterization check of Experiment B. The preprocessing contract is fixed within each comparison (pretrigger baseline subtraction, one-sided real FFT, Welch PSD floored at its first percentile, inverse-PSD weights, peak-aligned unit-amplitude template, fixed 60/20/20 train/validation/test split); all metrics are computed on held-out test events and reported as mean ± standard deviation over 8 seeds. Full simulation and preprocessing details are in Appendix G.

Metrics. Primary metrics are likelihood-aligned: the *weighted residual* $\chi^2 = \|x - \hat{x}\|^2_{\Sigma^{-1}}$ on test events (for fixed Σ, held-out Gaussian NLL is monotone in this quantity); the *energy-resolution proxy* σ_E, the standard deviation of the amplitude estimate on fixed-amplitude events, normalized by the CRB $\sigma_{\text{CRB}} = 1/\sqrt{N_\Phi}$; and *whitened subspace overlap*, the weighted cosine $\rho_w = |u_1^\dagger \Sigma^{-1} s_0| / (\|u_1\|_{\Sigma^{-1}} \|s_0\|_{\Sigma^{-1}})$ at rank 1 and principal angles for $k > 1$. The *raw held-out MSE* $\frac{1}{N}\sum_i \|x_i - \hat{x}_i\|_2^2$ is reported as a diagnostic: it is the objective optimized by isotropic PCA and standard linear AEs but is not the detector likelihood unless $\Sigma \propto I$.

7.2 Experiment A: rank-1 EMPCA reproduces the optimal filter

Question. Does rank-1 EMPCA reproduce the OF amplitude estimator on an internal detector calibration sample under matched preprocessing, as Theorem 4.1 predicts?

Setup. Internal K-α calibration traces with identical baseline correction, identical PSD weighting, and matched train/test conventions for both estimators. Because the sample is not a public benchmark, it is treated as a consistency check rather than as the primary evidence for the framework.

Metric. Weighted subspace cosine ρ_w between the learned rank-1 direction and the OF template; per-event amplitude agreement; residual-distribution agreement (KS test).

Result. Table 3 and Figure 2. The weighted cosine is $\rho_w = 0.999\,999\,97$, the median relative amplitude error is 9.7×10^{-5}, the amplitude correlation is $r = 0.9942$, and the residual KS test is consistent with identical fit quality ($p = 0.27$).

Table 3: Experiment A: OF vs. rank-1 EMPCA on the internal K-α calibration sample under matched preprocessing and weighting.

Metric	Value
$\max \lvert a_{\mathrm{OF}} - a_{\text{template GLS}} \rvert$	5.5339×10^{-4}
Weighted subspace cosine $\rho_w(U, s_0)$	0.9999999655
Amplitude correlation r	0.9942
Median relative error (origin fit)	9.6782×10^{-5}
Residual KS statistic D (p-value)	0.0303 (0.270)

Interpretation. Under matched preprocessing, whitening, weighting, and normalization, the learned rank-1 subspace coincides with the whitened template direction and the two estimators are functionally identical, as predicted at the population level by Theorem 4.1 and exactly by Theorem 4.2. Figure 3 shows the spectral-domain mechanism: for each noise color, the learned PC1 magnitude spectrum $|u_1(f)|^2$ is proportional to the SNR density $|S(f)|^2/J(f)$ to within numerical precision — rank-1 EMPCA recovers the whitened template direction for any stationary noise color.

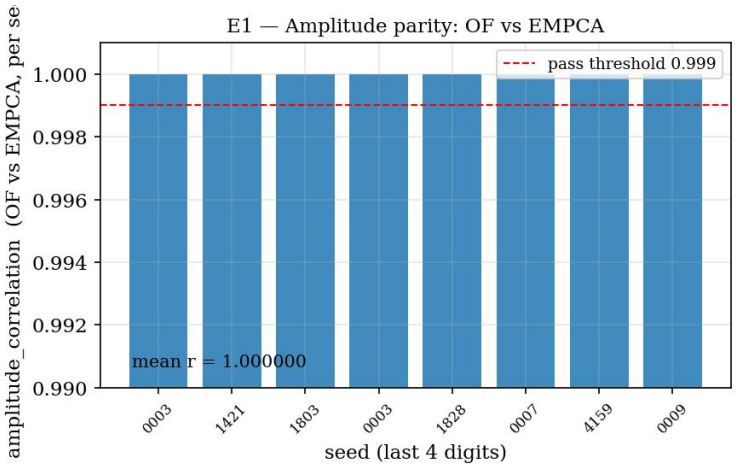


Figure 2: Experiment A: rank-1 EMPCA amplitudes against optimal-filter amplitudes on internal K-α calibration events under matched preprocessing and PSD weighting.

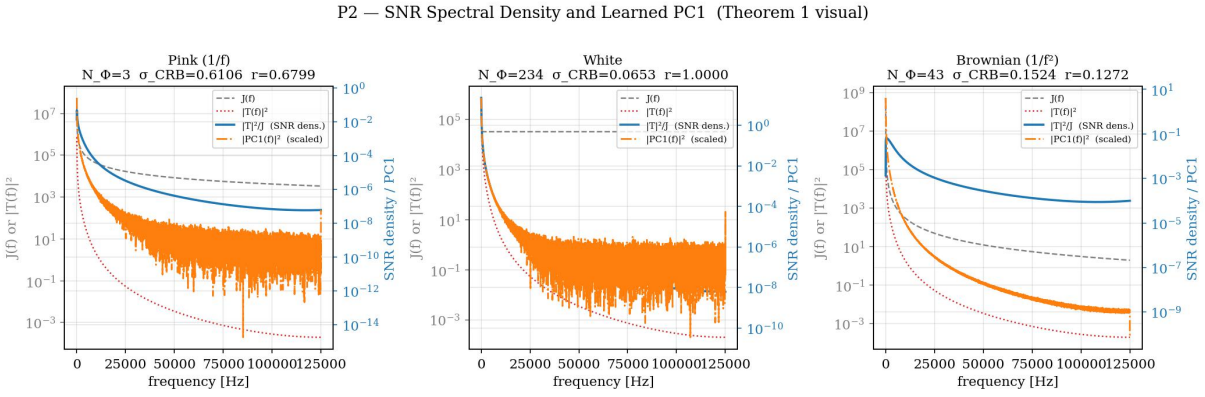


Figure 3: Experiment A, spectral view: SNR density $|S(f)|^2/J(f)$ (blue) and learned PC1 magnitude $|u_1(f)|^2$ (orange, scaled) for pink, white, and Brownian noise. N_Φ and the corresponding CRB are annotated per panel.

7.3 Experiment B: Bridge Theorem verification

Question. Does EMPCA converge to the exact minimizer of the noise-aware tied linear AE loss (27), as Theorem 5.1 predicts?

Setup. Simulated traces under all three noise colors; EMPCA at ranks $k = 1, 2, 3$ compared against the exact AE minimizer computed from the generalized eigenproblem. The internal K-α split is used as an additional parameterization check, not as the primary test.

Metric. Minimum principal-angle cosine between the two rank-k subspaces; residual distribution agreement.

Result. Figure 4 and Table 4. The minimum principal-angle cosine exceeds 0.9999 for all ranks and seeds (principal angle $\approx 0.005^\circ$ on the internal K-α split), and the residual distributions are indistinguishable (KS $p = 1.0$).

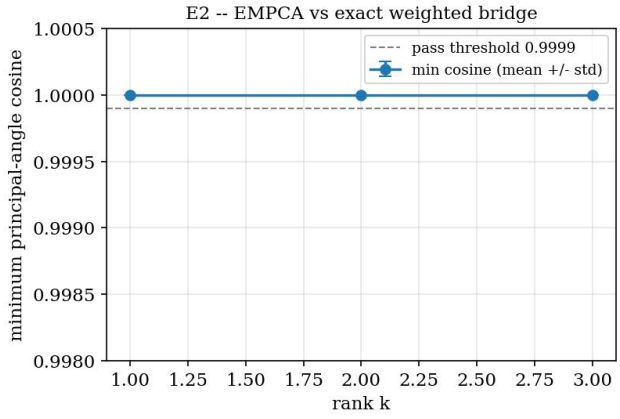


Figure 4: Experiment B: minimum principal-angle cosine between the EMPCA subspace and the exact noise-aware AE subspace versus rank k (mean $\pm$ std, 8 seeds).

Interpretation. The EMPCA algorithm and the noise-aware tied linear AE objective identify the same subspace, confirming that the encoder–decoder formulation is a reparameterization, not a different estimator. Comparing complex EMPCA to a real-feature rank-k model instead produces apparent mismatch; this is a representation mismatch, not a violation of the theorem.

Table 4: Experiment B: EMPCA vs. exact noise-aware linear AE on the internal K-α split (complex-whitened parameterization check).

Metric	Value
Principal-angle cosine (EMPCA vs. exact AE)	0.9999999967
Principal angle	0.0046°
Residual KS statistic D (p-value)	0.0014 (1.0)
Relative mean residual difference	8.2×10^{-7}

7.4 Experiment C: white-noise control

Question. When $\Sigma \propto I$, do isotropic PCA and noise-aware EMPCA agree, as the likelihood argument requires?

Setup. Identical rank, training events, preprocessing, and train/test split for both methods; white noise.

Metric. Difference in held-out weighted residual; whitened subspace angle θ_w between the two learned bases.

Result. First rows of Table 5 and the white-noise bar of Figure 5: the residual difference is consistent with zero and $\theta_w = 0^\circ$ — the two methods learn the same direction.

Interpretation. Under white noise the two objectives are the same likelihood up to scale, so agreement is required, and observed. This control isolates loss geometry as the causal variable in Experiment D: rank, optimizer, data, and preprocessing are identical across the comparison.

7.5 Experiment D: colored noise and metric reversal

Question. Under colored noise, do isotropic MSE and the Σ^{-1}-weighted likelihood prefer different reconstructions, and when is the MSE preference misaligned with the assumed detector likelihood?

Setup. Same controlled comparison as Experiment C under pink and Brownian noise: isotropic PCA versus noise-aware EMPCA at fixed rank, identical data and preprocessing ($N_{\text{train}} = 400$, $N_{\text{test}} = 200$, 8 seeds).

Metric. Raw held-out MSE, held-out weighted residual χ^2, whitened subspace angle θ_w, and the energy-resolution proxy σ_E.

Result. Table 5 and Figures 5–6. Under Brownian noise, isotropic PCA achieves 45.1% *lower* raw MSE while incurring a 0.24% *higher* weighted residual, and the two learned subspaces are separated by $\theta_w = 87.1°$ in the whitened metric; pink noise shows the same pattern (+6.8% MSE advantage, −0.23% χ^2 disadvantage, $\theta_w = 87.0°$). The weighted objective also improves σ_E whenever the PSD is colored (Figure 6). The rank sweep in Appendix F reports the same ordering across the tested ranks; its small numerical differences should not be interpreted as a general monotonicity claim.

Table 5: Experiment D (metric reversal): isotropic PCA vs. noise-aware EMPCA at rank $k = 1$ on identical data, preprocessing, and held-out split. ΔMSE is the relative raw-MSE difference (positive = isotropic PCA better on raw reconstruction); $\Delta\chi^2$ is the relative weighted-residual difference (negative = EMPCA better on detector likelihood); θ_w is the whitened angle between the learned bases. Mean over 8 seeds.

Noise type	ΔMSE (iso. PCA advantage)	$\Delta\chi^2$ (EMPCA advantage)	θ_w
White	0.0%	0.00%	0.0°
Pink	+6.8%	−0.23%	87.0°
Brownian	+45.1%	−0.24%	87.1°

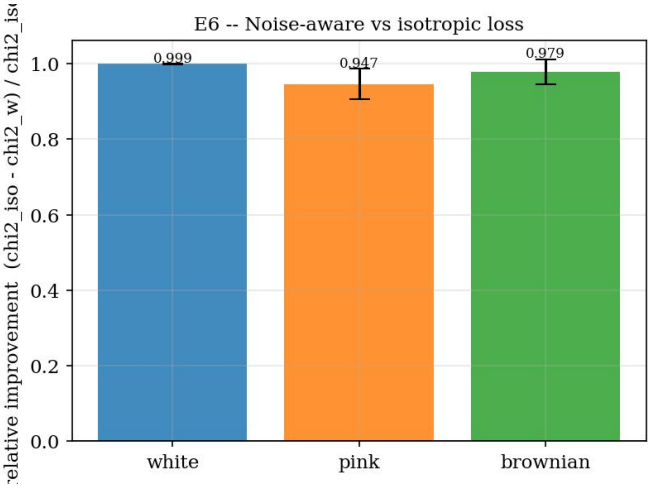


Figure 5: Experiment C/D: relative improvement in held-out weighted residual of noise-aware EMPCA over isotropic PCA for three noise types (mean ± std, 8 seeds). The improvement vanishes for white noise and appears under colored noise, as the likelihood argument predicts.

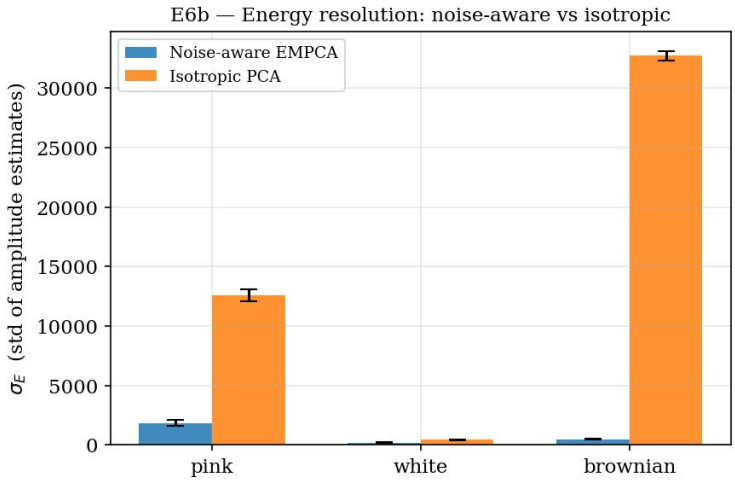


Figure 6: Experiment D in energy-resolution units: σ_E for noise-aware EMPCA and isotropic PCA. The gain appears in colored-noise regimes and vanishes for white noise.

Interpretation. This is the metric reversal of Section 6 realized empirically. Isotropic PCA concentrates on high-variance, noise-dominated (low-frequency) directions because that is what its objective rewards; EMPCA concentrates on high-SNR directions weighted by Σ^{-1}. The small magnitude of the χ^2 gap understates the difference: the learned representations are nearly orthogonal in the whitened metric, so the two losses are selecting substantially different representation directions, not fine-tuning the same solution. Model selection by raw MSE under colored noise would prefer a subspace misaligned with the assumed detector likelihood. An oracle comparison on noiseless signals is intentionally omitted: the linear performance bound is already given analytically by the CRB (Section 3.2).

Scope note. The loss-geometry comparison is simulation-based and is the primary validation because the true covariance and signal rank are known. The internal K-α sample checks implementation consistency only. A domain-transfer claim requires an independently reproducible public dataset, a noise estimate derived without using the fitted trace, and the same predeclared metrics; such experiments must be completed before they are reported as validation rather than future work.

7.6 Experiment E: controlled validation on multidimensional pulse families

Question. Does the rank-k extension matter for detector-relevant quantities on controlled pulse families whose intrinsic dimension is known?

Setup. The multi-dimensional simulated family with timing jitter (a genuinely two-dimensional signal subspace: mean pulse plus derivative mode), under all three noise colors.

Metric. Energy resolution $\sigma_E(k)$ normalized to the OF baseline; amplitude bias under large timing jitter; held-out weighted residual versus rank.

Result. Figure 7. At $k = 1$ the resolution matches the OF baseline within 2% for all noise types, as required by the equivalence theorems. For the jittered family, rank-2 EMPCA improves resolution and then saturates — consistent with the true two-dimensional signal subspace — and absorbs the timing-jitter mode into PC2, reducing the amplitude bias by more than 30% for all noise types. Residual goodness-of-fit becomes statistically consistent with the noise-only null exactly at the true rank (KS $p > 0.05$ at $k = 2$). Supporting figures (rank saturation, timing bias, residual calibration, PC interpretation with bootstrap CIs) are in Appendix F.

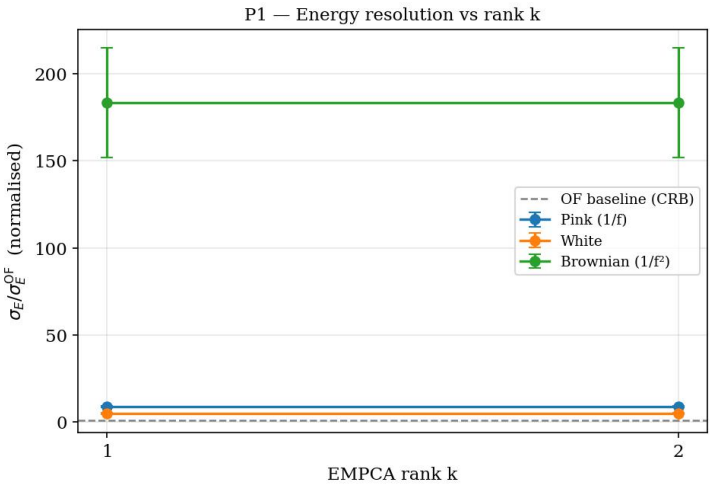


Figure 7: Experiment E: energy resolution $\sigma_E(k)/\sigma_E^{\mathrm{OF}}$ versus EMPCA rank k for pink, white, and Brownian noise (mean $\pm$ std, 8 seeds); dashed line = OF baseline. All noise types match the OF baseline at $k = 1$ and improve at $k = 2$ in the timing-jitter regime, saturating beyond.

Interpretation. The hierarchy behaves as the theory prescribes in a controlled detector-motivated setting: OF degeneracy at rank 1, physically interpretable gains when the signal family has additional dimensions, and calibrated residuals at the true rank. This supports using the noise-aware linear hierarchy as a baseline for subsequent public-domain and detector-specific validation; it does not by itself establish domain transfer.

7.7 Evidence-bundle synthetic gates S1–S9

Question. Do the controlled validation gates pass their predeclared multi-seed criteria when run as the paper-facing evidence bundle?

Setup. The transferred synthetic suite swept gates S1–S9 over ten deterministic seeds, with held-out evaluation where required. The gates cover optimal-filter normalization, weighted-subspace geometry, the independent linear-AE/EMPCA bridge, the white-noise PCA control, colored-noise metric behavior, timing/rank sensitivity, covariance-estimation convergence, residual calibration, and multichannel covariance.

Metric. Each gate uses its predeclared headline metric and interval summary from `synthetic_gate_summary.csv`. The metrics are interpreted only in their controlled synthetic settings.

Result. All nine gates passed their multi-seed regression criteria (Table 6 and Figure 8).

Table 6: Completed synthetic validation gates from the transferred evidence bundle. Values are mean and 95% interval for the predeclared headline metric unless the metric is a paired gate.

Gate	Headline metric	Mean [95% interval]	Status
S1	`sigma_over_crb`	$0.994\,[0.985, 1.002]$	passed
S2	`weighted_angle_deg`	$3.862\,[3.803, 3.924]$	passed
S3	`relative_gap`	$1.95\times10^{-16}\,[0.92, 2.99]\times10^{-16}$	passed
S4	`max_principal_angle_deg`	$1.48\times10^{-6}\,[1.26, 1.71]\times10^{-6}$	passed
S5	`pca_weighted_residual_to_observed`	$48.49\,[48.35, 48.64]$	passed
S6	`best_rank_by_clean_mse`	$4.6\,[4.2, 5.0]$	passed
S7	`sigma_over_oracle` at largest calibration size	$1.032\,[1.025, 1.040]$	passed
S8	`mean_chi2_per_dof`	$0.997\,[0.995, 1.000]$	passed
S9	`diagonal_over_full_sigma`	$1.025\,[1.020, 1.030]$	passed

Interpretation. The synthetic ladder supports the controlled mathematical and software validation claims. It does not establish calibration on nonstationary real detector noise, and it is therefore separated from the GWOSC stress test below.

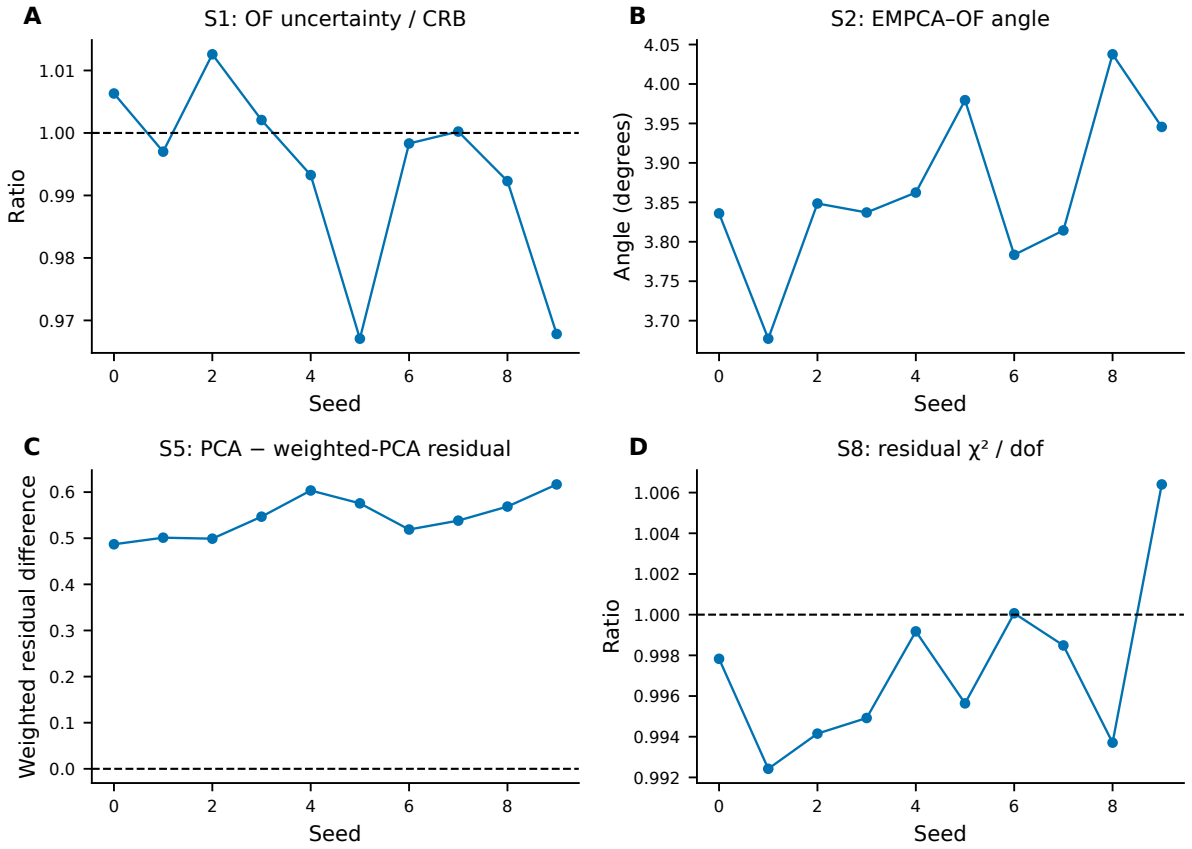


Figure 8: Representative multi-seed synthetic validation gates. A, The empirical optimal-filter amplitude standard deviation divided by the Cramér–Rao bound across ten seeds; the dashed line marks unity. B, The weighted principal angle between the EMPCA direction and the optimal-filter direction. C, The held-out weighted-residual difference between ordinary PCA and weighted PCA in colored noise; positive values favor weighted PCA. D, Mean whitened residual chi-square per degree of freedom; the dashed line marks unity. Each point is one deterministic seed.

7.8 Remote reproducibility and GWOSC data integrity

Question. Was the public-data stress-test workflow run reproducibly, and did the selected GWOSC interval have the required data coverage?

Setup. The selected archived run, `20260622T175125Z_b169c1f595a4`, used 256 seconds of public GW150914-centered H1 and L1 strain sampled at 4096 Hz. The workflow records the environment, dependencies, test output, core experiment execution, table generation, figure generation, and checksums.

Result. Stage 0 remote reproducibility passed on a clean Linux checkout. Official `H1_DATA` and `L1_DATA` coverage was present, the event window was valid for both detectors, all 62 candidate off-source windows were valid for both detectors, and no evaluation windows were flagged by the recorded broad data-quality diagnostics. The raw strain arrays are not stored in this writing bundle; the bundle stores metadata, checksums, configs, JSON evidence, derived tables, and figures.

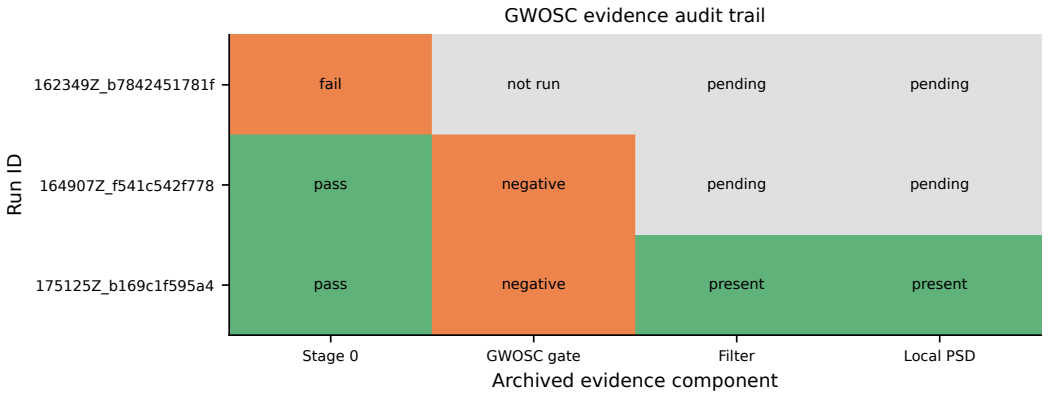


Figure 9: Archived remote-run audit trail. Rows are synchronized GWOSC evidence runs and columns show Stage 0 reproducibility, the baseline GWOSC scientific gate, and the presence or absence of the follow-up filtering and local-PSD evidence. This figure separates operational success, scientific acceptance, negative results, and pending evidence.

Interpretation. The remote run supports an operational reproducibility statement and a data integrity statement for this interval. It does not by itself imply that the real-data scientific gate passed.

7.9 GWOSC PSD reference and score-path diagnostics

Question. Does the repository PSD estimator match an independent implementation on the same calibration windows?

Setup. The repository Hann-windowed, constant-detrended, bias-corrected median Welch PSD was compared with GWpy's median-Welch PSD on identical H1 and L1 calibration windows from the selected interval.

Result. For both detectors, the repository/GWpy PSD ratio had median, fifth percentile, and ninety-fifth percentile equal to 1.0 at recorded precision; the relative L^2 error was 0.0, and the maximum absolute $\log_{10}$ ratio was 0.0. The repository generalized least-squares score was strongly correlated with the repository direct-whitened score (0.974 for H1 and 0.997 for L1). Its correlation with the earlier GWpy inverse-spectrum-FIR score was lower and detector-dependent (0.103 for H1 and 0.729 for L1). Because those score paths did not compute an identical finite-filter statistic, this latter comparison is diagnostic only.

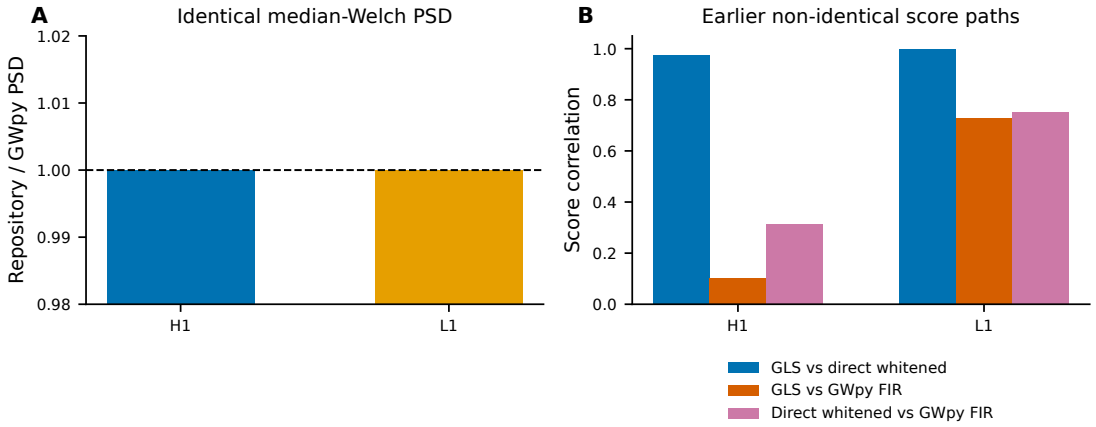


Figure 10: Independent reference and score-path diagnostics. A, The repository Hann-windowed, bias-corrected median PSD divided by GWpy's PSD on identical calibration windows. Both detector ratios equal unity at recorded precision. B, Correlations among the repository generalized least-squares score, the repository direct-rFFT-whitened score, and the earlier GWpy inverse-spectrum-FIR score. Panel B compares non-identical effective filters and is diagnostic rather than an implementation-equivalence acceptance test.

Interpretation. The PSD estimator was independently verified for the same estimator and windows. This is a necessary implementation check, but it is not sufficient for

calibrated likelihood inference on held-out real detector noise.

7.10 GWOSC shared-FIR implementation identity

Question. Do the explicit FFT-convolution path and the GWpy convolution path agree when they are forced to compute the same finite FIR statistic with identical coefficients and score normalization?

Setup. The predeclared shared-FIR comparison applied identical GWpy-designed FIR coefficients through both explicit FFT convolution and GWpy convolution over the FIR-duration and edge-trim grid. This experiment separates the narrow software-identity question from the methodological sensitivity between the original PSD-domain generalized least-squares statistic and a finite-duration FIR statistic.

Result. The shared-statistic implementation identity gate passed. The maximum absolute score differences were 4.98×10^{-15} for the stationary synthetic control, 1.60×10^{-12} for H1, and 1.45×10^{-12} for L1, all below the 10^{-10} threshold.

Table 7: GWOSC shared-FIR implementation-identity follow-up. Explicit FFT convolution and GWpy convolution used identical FIR coefficients and score normalization.

Dataset	Maximum absolute score difference	Gate status
Synthetic control	4.98×10^{-15}	passed
H1 real window	1.60×10^{-12}	passed
L1 real window	1.45×10^{-12}	passed

Interpretation. The result resolves the shared-statistic software-identity question: two implementations agree when they compute the same FIR statistic. It does not prove that the original PSD-domain GLS statistic and a finite-duration FIR statistic are mathematically equivalent; differences between those definitions remain diagnostic sensitivity.

7.11 GWOSC global-PSD held-out null calibration

Question. Does a globally estimated PSD predict the held-out null matched-statistic spread on the selected real GWOSC interval?

Setup. The predeclared real-data gate evaluated deterministic random splits and contiguous chronological blocks for H1 and L1. The primary diagnostic is the ratio of the observed held-out amplitude standard deviation to the PSD-derived prediction.

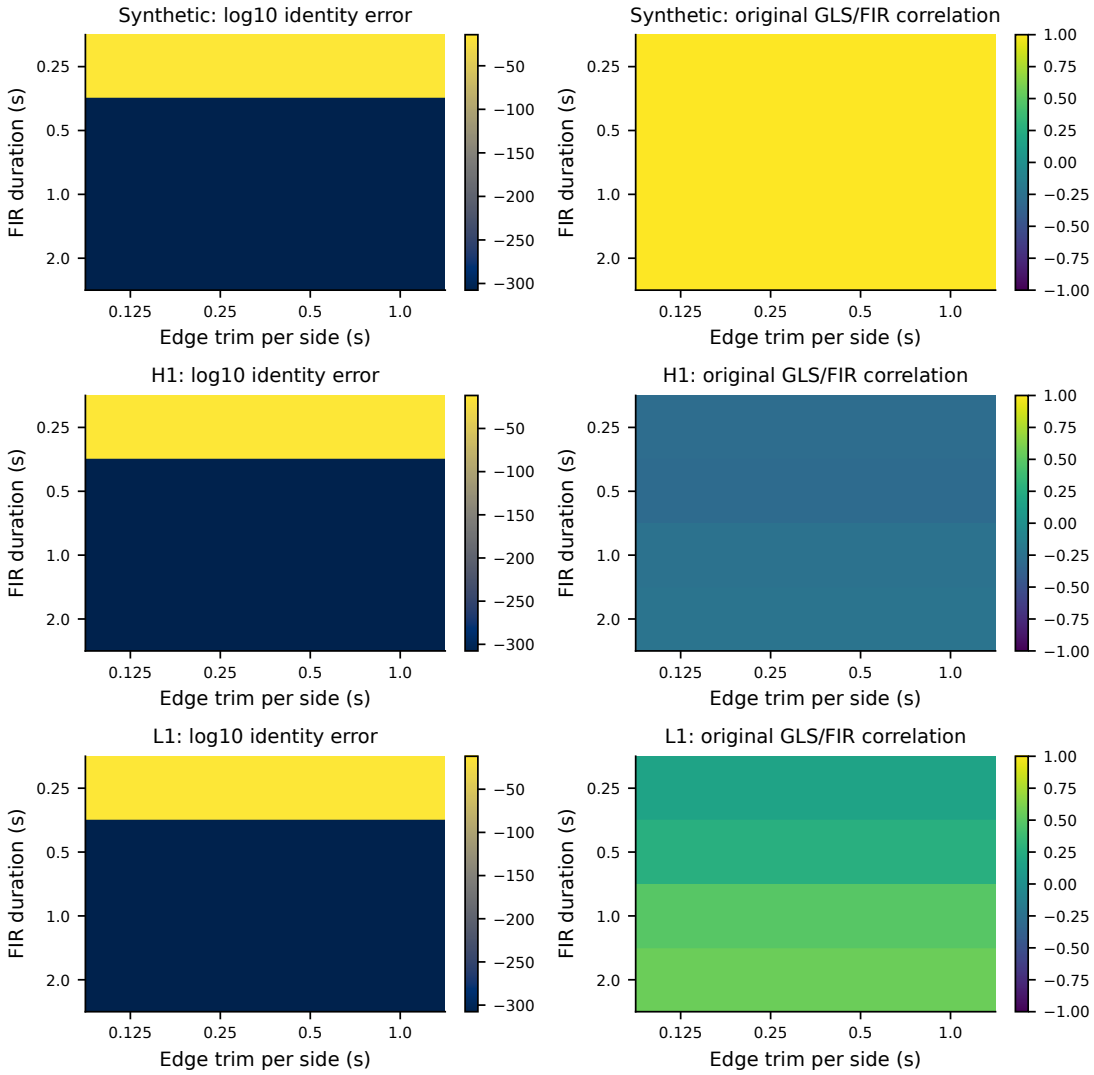


Figure 11: Shared-FIR implementation identity over the predeclared FIR-duration/edge-trim grid. For stationary synthetic data and H1/L1 real windows, the left panels show the maximum absolute score difference between explicit FFT convolution and GWpy convolution when both paths use identical FIR coefficients and score normalization. The right panels show the original PSD-domain GLS/shared-FIR correlation as a methodological sensitivity diagnostic. The identity gate passed; the GLS/FIR comparison is not a mathematical-equivalence test for the original statistic and the finite FIR statistic.

Result. The global-PSD model failed the predeclared real-noise acceptance gate (Table 8 and Figure 12). In random splits, H1 had median/minimum/maximum observed-to-predicted ratios 1.986/1.396/7.166, and L1 had 1.263/1.209/3.093. In chronological blocks, H1 had 3.183/1.549/10.480, and L1 had 2.964/1.973/7.619. The archived event scores were 6.636 for H1 and −0.289 for L1, and the empirical injection SNRs were 2.396 for H1 and 6.435 for L1; these values are retained only as uncalibrated diagnostics because the corresponding real-noise calibration gates do not pass.

Table 8: GWOSC held-out null calibration summary for the selected GW150914-centered interval. Ratios greater than one indicate observed held-out spread larger than the PSD-derived prediction.

Detector and split	Median/minimum/maximum ratio	Gate status
H1 random splits	1.986/1.396/7.166	failed
L1 random splits	1.263/1.209/3.093	failed
H1 chronological blocks	3.183/1.549/10.480	failed
L1 chronological blocks	2.964/1.973/7.619	failed

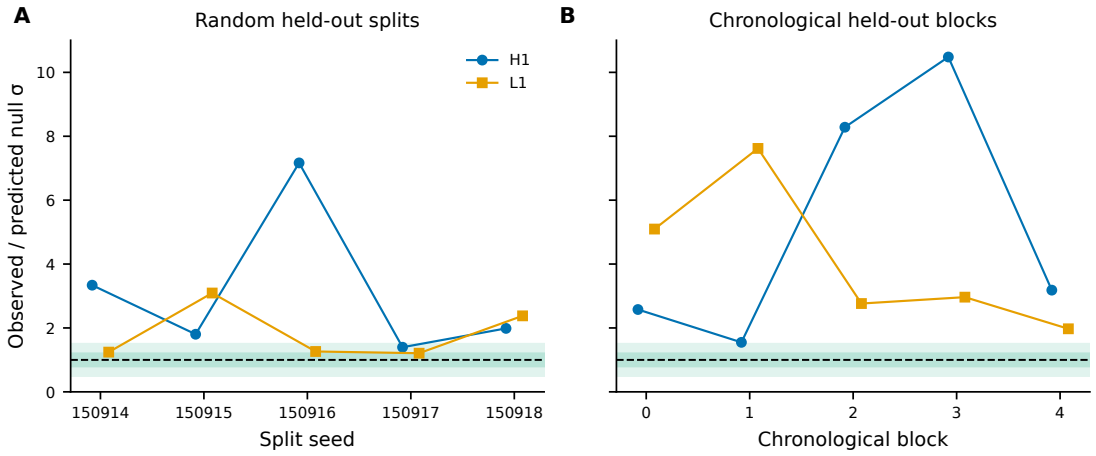


Figure 12: Held-out amplitude-spread calibration for 256 seconds of GW150914-centered H1 and L1 open data. A, Five deterministic random calibration/evaluation splits. B, Five contiguous chronological evaluation blocks with nearby calibration windows. The ordinate is the observed held-out amplitude standard deviation divided by the PSD-derived prediction. The dashed line marks unity, the darker green band marks the predeclared median interval $[0.8, 1.2]$, and the lighter green band marks the per-split interval $[0.5, 1.5]$. The predeclared gate failed for both detectors and both split designs.

Interpretation. The current public-data result is negative: exact PSD-estimator agreement with GWpy did not yield calibrated held-out template-score spread for a global PSD model on this interval. The result is interval-specific and does not identify a unique cause. Plausible explanations include local spectral drift, template-projected spectral variation, narrow-band features, transient non-Gaussianity, and statistic-definition differences; the completed follow-ups above and below narrow the interpretation but do not turn this real-data gate into a pass.

7.12 GWOSC time-local PSD follow-up

Question. Does the predeclared primary 64-second local PSD model calibrate the held-out template-score spread on the same real GWOSC interval?

Setup. The time-local PSD follow-up compared a leave-one-out global PSD with fixed 32-, 64-, and 96-second local calibration radii. The 64-second local model was the predeclared primary model. A stationary synthetic control was run through the same local-PSD diagnostic to test for generic normalization errors.

Result. The stationary synthetic control passed, with primary local-64 score standard deviation 1.148. The real-data primary local-64 gate failed for both detectors despite full coverage: H1 had global/local-64 score standard deviations 1.771/9.424, and L1 had 7.288/8.827, compared with the predeclared real-data target interval $[0.8, 1.2]$. Coverage was 1.0 for both H1 and L1.

Table 9: GWOSC time-local PSD follow-up. The 64-second local PSD model was the predeclared primary model; the real-data acceptance interval was $[0.8, 1.2]$.

Dataset	Global score std	Local-64 score std	Gate status
Synthetic control	—	1.148	passed
H1 real data	1.771	9.424	failed
L1 real data	7.288	8.827	failed

Interpretation. The local-PSD follow-up is a negative real-data result. Because the stationary synthetic control passed, the failure is best interpreted as real-noise/model inadequacy for this interval rather than a generic normalization error. The GWOSC stress test therefore passed PSD/reference and shared-FIR implementation checks but failed held-out real-noise calibration under both global and predeclared local PSD models.

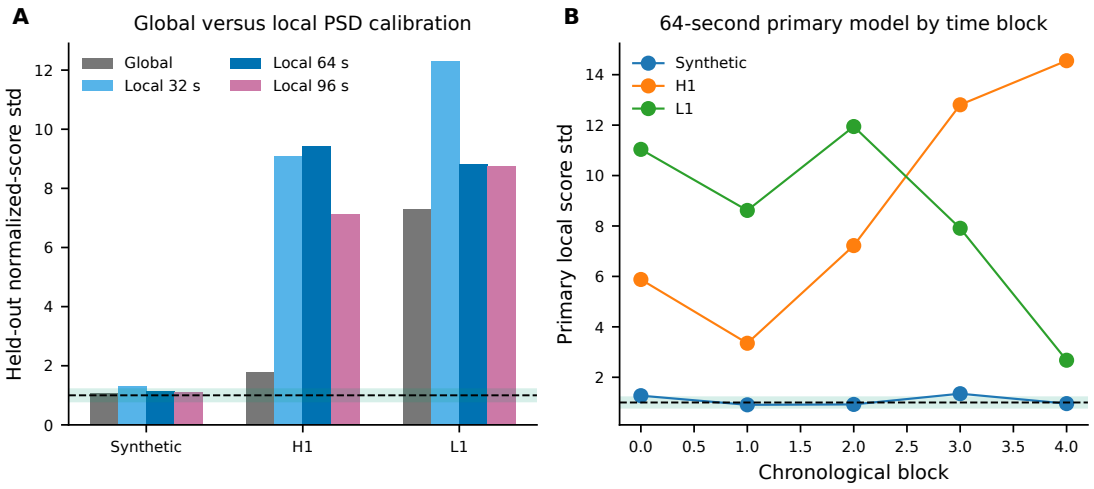


Figure 13: Negative time-local PSD follow-up. A, Normalized held-out score standard deviations for leave-one-out global PSDs and fixed 32-, 64-, and 96-second local PSDs. The 64-second model is the predeclared primary model. The stationary synthetic control passed, but H1 and L1 real-data score dispersion remained far outside the $[0.8, 1.2]$ acceptance band. B, Five chronological blocks for the primary model, showing persistent real-data instability. Template-projected and fixed narrow-band power ratios are archived as diagnostics and are not used to define post hoc frequency cuts.

Event and injection scores are diagnostic only, and no calibrated GWOSC significance or sensitivity claim is made.

7.13 Future and diagnostic-only result slots

The following entries remain working-draft placeholders or diagnostic-only slots. They are not completed positive results and must not be converted into validation claims before new evidence is synchronized and reviewed.

Confirmatory GWOSC interval. The present GWOSC analysis is a stress test on one public 256-second interval around GW150914 and should be treated as interval-specific.

[PLACEHOLDER — pending confirmatory GWOSC evidence. Insert confirmatory held-out real-noise calibration results only after an untouched interval/event set is run under the frozen protocol and synchronized into `transfer_paper`.]

CRESST/SCRESST validation. CRESST/SCRESST is the planned detector-pulse real-data validation path, because its schema and preprocessing assumptions still need independent review before it can support a positive real-data claim.

[PLACEHOLDER — pending CRESST evidence. Insert CRESST/SCRESST validation only after the data schema, preprocessing, calibration split, experiment outputs, and acceptance review are synchronized into `transfer_paper`.]

Calibrated event significance and injection sensitivity. The archived event and injection scores are retained as diagnostics only, because the corresponding held-out real-noise calibration gates failed.

[PLACEHOLDER — pending calibrated event/sensitivity evidence. Insert GW150914 significance or injection-sensitivity claims only after the real-noise null calibration gate passes and the event/injection scoring protocol is frozen, rerun, synchronized, and reviewed.]

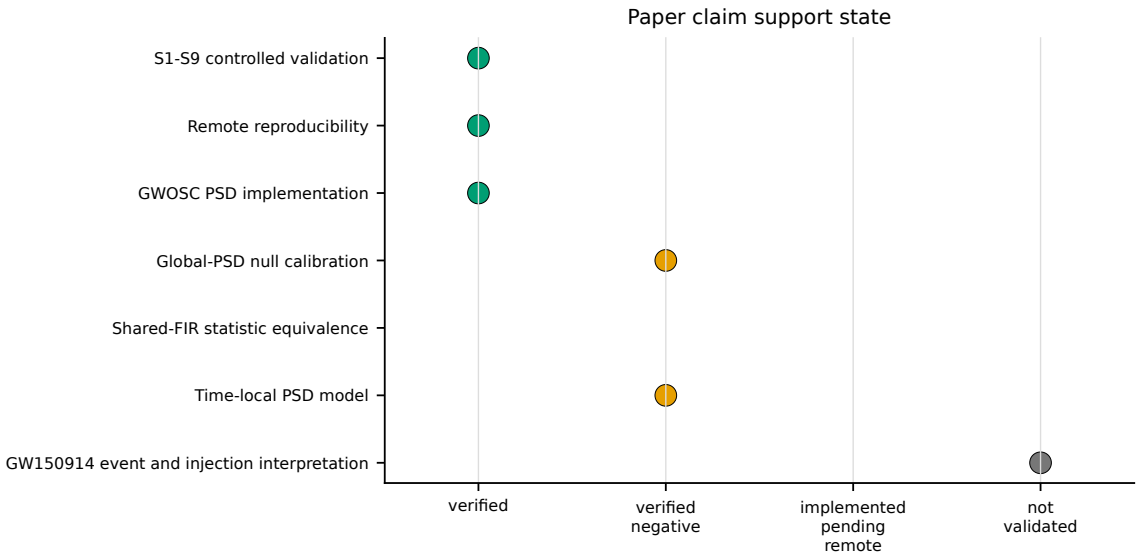


Figure 14: Claim-state matrix used as a writing-control figure. Each row is a paper claim area and the marker indicates whether the current bundle supports it as verified, verified-negative, implemented-but-pending, or not validated. This plot is not a scientific result by itself; it separates supported manuscript statements from overclaims.

8 Discussion

Likelihood geometry as the organizing principle. The estimator hierarchy of Table 1 rests on a single principle: a reconstruction method is statistically efficient when its objective respects the geometry imposed by the noise covariance. The central object is therefore not any particular algorithm but the noise-induced metric Σ^{-1}. Changing the architecture while keeping an isotropic loss produces a more expressive but still misspecified model; changing the loss to encode Σ^{-1} while keeping a simple linear architecture recovers the maximum-likelihood geometry. This asymmetry — loss geometry first, architecture second — is the transferable lesson of the linear theory, and Experiment D realizes it concretely: with everything else held fixed, the loss alone determines whether the learned subspace aligns with detector-relevant signal-to-noise structure or with raw variance.

One hierarchy, three communities. Viewing OF, EMPCA, and the noise-aware tied linear AE as one hierarchy of constrained ML projections has practical value beyond conceptual economy. It tells the detector physicist exactly when a learned subspace can improve on the optimal filter within the nested-subspace model (when the signal family contains directions outside the fixed template) and how to quantify that improvement

through the weighted projection residual. It tells the statistician that EMPCA is not merely an algorithm but the constrained ML estimator for general Σ, the heteroskedastic generalization of probabilistic PCA [9] beyond factor analysis's diagonal noise [10]. And it tells the machine-learning practitioner that the tied linear AE under the weighted loss is the encoder–decoder parameterization of the same problem — so a nonlinear AE that fails to beat this baseline on likelihood-aligned metrics has not demonstrated a gain from its additional expressivity in that comparison.

Implications for detector ML. In scientific instrument settings — calorimeters, gravitational-wave detectors, radio telescopes, imaging arrays — the noise model is often well characterized from calibration data, yet reconstruction networks are routinely trained with isotropic MSE. The framework makes the cost explicit: under structured noise, isotropic MSE biases the learned representation toward variance-dominant rather than SNR-dominant directions, and model selection by raw MSE can prefer a reconstruction misaligned with the assumed detector likelihood (the metric reversal of Experiment D). Replacing MSE by the Σ^{-1}-weighted loss requires no architectural change and is a natural starting point when the Gaussian covariance model is adequate. The same reasoning fixes the evaluation protocol: nonlinear reconstruction models should be benchmarked on held-out likelihood, weighted residual, and detector-relevant resolution — not raw Euclidean error alone.

Real-data stress test and negative calibration result. The GWOSC analysis separates implementation correctness from statistical adequacy. On the selected 256-second GW150914-centered interval, the repository Hann-windowed, bias-corrected median PSD matched GWpy exactly on identical calibration windows, so the failed real-data gate is not explained by a simple disagreement with that reference PSD estimator. A shared-FIR follow-up further showed that explicit FFT convolution and GWpy convolution agree when both paths use the same FIR coefficients and score normalization. Nevertheless, the globally estimated PSD did not predict held-out template-score spread in random or chronological splits for H1 and L1, and the predeclared primary 64-second local PSD model also failed the real-data acceptance gate for both detectors. The stationary synthetic control for the local-PSD experiment passed, so the failure is best interpreted as real-noise/model inadequacy for this interval rather than a generic normalization error. This is a negative public real-noise stress test and a boundary of the present validation, not an event-detection claim.

Relation to neighboring model families. The framework is the fully general-covariance member of a family of Gaussian latent-variable models: probabilistic PCA

assumes $\Sigma = \sigma^2 I$, factor analysis allows diagonal Σ, and the present theory allows arbitrary $\Sigma \succ 0$, which is essential for correlated time bins and channels. A linear VAE with decoder likelihood $\mathcal{N}(Wz, \Sigma)$ reduces to the noise-aware reconstruction loss in the deterministic-encoder limit [24]; adding the KL term would give a heteroskedastic linear VAE, which we do not develop here. Denoising autoencoders trained under Gaussian corruption recover the same subspace when the corruption covariance equals the measurement covariance [16, 17]; in that reading, the noise-aware AE is a denoising AE whose corruption is the physical measurement noise. Relative to physics-informed ML [19, 20], the distinction is that Σ is not a physics penalty added to a standard loss — it is the likelihood itself.

What this paper does not solve. The theory is exact only in the linear Gaussian regime. It does not establish how much of the loss-geometry advantage survives for nonlinear signal manifolds, non-Gaussian noise, or learned covariance models; it does not provide a nonlinear architecture; and it does not address online or streaming estimation. These boundaries are stated precisely below and define the companion nonlinear program: replace the linear encoder by a nonlinear map while retaining the Σ^{-1}-weighted likelihood, and benchmark against the linear hierarchy developed here.

Architecture and coverage belong to a separate question. The likelihood geometry specifies how residuals should be weighted, but it does not determine whether a training sample excites every physically relevant mode or which inductive biases select among expressive nonlinear solutions. A companion study will examine how architecture constraints and training-distribution coverage interact with the likelihood geometry. Those questions are intentionally not developed here because they require different experiments and claims from the linear theory and controlled validation in this paper.

9 Limitations

The theory is linear. All equivalence results are statements about linear subspace models. The true manifold of shifted, position-dependent, multi-mode pulses is nonlinear; the linear theory is exact only insofar as the signal family is well approximated by a k-dimensional subspace.

Gaussian likelihood. The χ^2 objective is the negative log-likelihood only for Gaussian noise. Under heavy-tailed contamination (vibration transients, cosmic-ray glitches, pile-up), the Gaussian-weighted estimator is generally outlier-sensitive and need not be

efficient or consistent for the desired signal parameter; robust M-estimation breaks the closed-form EM structure and is not developed here.

Covariance must be known or estimated. Every estimator in the hierarchy requires Σ (equivalently the noise PSD) from dedicated noise data. Estimation error or non-stationarity (drifting baselines, temperature dependence) degrades the whitening and weakens the optimality guarantees; robust and adaptive covariance estimation is outside the present scope.

The current GWOSC evidence is interval-specific. The public-data evidence covers one 256-second interval around GW150914. Split seeds and chronological blocks within that interval are useful stress tests, but they are not independent astrophysical replications and do not support a general GWOSC calibration claim. The archived event and injection scores are uncalibrated diagnostics because the held-out real-noise calibration gates failed under both global and predeclared local PSD models. Confirmatory intervals or event sets must be run under a frozen protocol before any broader real-noise calibration, event-significance, or injection-sensitivity claim can be made.

Implementation checks are not calibration. The GWOSC follow-ups narrow what failed without converting the stress test into a positive validation. The shared-FIR result proves identity between two implementations of the same finite FIR statistic; it does not prove that the original PSD-domain GLS statistic and a finite-duration FIR statistic are mathematically equivalent. The time-local PSD experiment passed its stationary synthetic control but failed H1 and L1 real-data calibration for the predeclared primary 64-second model. Correct PSD implementation and shared-statistic software identity are therefore not sufficient for calibrated inference on this real-noise interval.

Real detector noise may require richer background models. The failed GWOSC gates point to missing real-noise structure, not to a passed detector validation. Future GWOSC work should be a new predeclared Stage 3b with a larger independent background, stricter veto/category handling, explicit outlier or glitch models, robust or shrinkage PSD estimation, and frozen frequency-window and threshold choices. Positive real-data validation should come from a separately predeclared detector-pulse dataset such as CRESST/SCRESST, after schema, preprocessing, split, and acceptance criteria are reviewed.

Linear subspaces, not signal manifolds. EMPCA and the noise-aware linear AE learn linear subspaces. They approximate, but cannot represent, curved signal manifolds

such as the full time-shift family at large jitter or saturating detector response.

No nonlinear architecture is provided. This paper deliberately stops at the linear baseline. It makes no claims about convolutional, recurrent, or transformer reconstruction models beyond the recommendation that their likelihood assumptions be stated explicitly and that they be evaluated against this baseline.

Future work. The natural extensions are nonlinear manifold models trained under the Σ^{-1}-weighted likelihood, non-Gaussian likelihoods for robust reconstruction, geometry-aware uncertainty quantification (e.g. a heteroskedastic linear VAE), a new predeclared GWOSC Stage 3b for real-noise background modelling, CRESST/SCRESST validation after schema and preprocessing review, and online variants for real-time triggering. Architecture and training-coverage studies are reserved for the companion paper described above.

10 Conclusion

Signal reconstruction is not only an architecture problem. The likelihood defines the geometry of the problem, the signal model defines the admissible reconstructions, and the architecture merely parameterizes the solution. In the linear Gaussian regime this separation is exact: the detector covariance induces the Σ^{-1} metric on observation space, and optimal filtering, EMPCA, and the noise-aware tied linear autoencoder form one hierarchy of constrained maximum-likelihood projections — a fixed rank-1 template, its structured extensions, and a learned rank-k subspace, related by exact degeneracies (Theorems 4.1, 4.2 and 5.1). The practical consequence is a loss-design rule: isotropic MSE silently assumes white noise, and under colored detector noise it can select a representation nearly orthogonal to the statistically likelihood-aligned one. When a Gaussian noise model is adequate and its covariance is measurable, that covariance belongs in the training and evaluation objective. The completed GWOSC stress test adds a practical caveat: PSD-estimator correctness and shared-FIR implementation identity did not produce calibrated real-noise inference under either the global PSD model or the predeclared primary 64-second local PSD model on the selected interval. The linear theory developed here is therefore offered as a baseline geometry against which more expressive scientific reconstruction models can be measured, with real-data calibration treated as an empirical gate rather than an assumption.

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A Detailed Optimal-Filter Derivations and the Cramér–Rao Bound

This appendix expands the summary statements of Section 3.2: unbiasedness, variance, Fisher information, and the energy-resolution formula for the OF estimator.

Unbiasedness. Taking the expectation of $\hat{A} = \sum_k \Phi_k^{(0)} X_k$ under the model $X_k = AS_k + N_k$:

$$\mathbb{E}[\hat{A}] = \sum_k \Phi_k^{(0)} A S_k = A \sum_k \frac{S_k^* S_k}{J_k N_\Phi} = A \cdot \frac{N_\Phi}{N_\Phi} = A. \tag{30}$$

Estimator variance. Because the noise bins N_k are independent with $\mathbb{E}[|N_k|^2] = J_k$,

$$\mathrm{Var}(\hat{A}) = \sum_k |\Phi_k^{(0)}|^2 J_k = \sum_k \frac{|S_k|^2}{J_k^2 N_\Phi^2} J_k = \frac{1}{N_\Phi^2} \sum_k \frac{|S_k|^2}{J_k} = \frac{1}{N_\Phi}, \tag{31}$$

which is Eq. (14) of the main text.

Fisher information and the CRB. The Fisher information for the scalar parameter A from a single trace is

$$\mathcal{I}(A) = -\mathbb{E}\left[\frac{\partial^2 \log \mathcal{L}}{\partial A^2}\right] = 2 \sum_k \frac{|S_k|^2}{J_k} = 2N_\Phi, \tag{32}$$

where the factor of 2 arises from the real–imaginary decomposition under the complex Gaussian model (and is absent in a real-valued model; conventions can also differ by a factor of 2 between one-sided and two-sided PSDs — see Appendix G). The CRB is $\mathrm{Var}(\hat{A}) \geq 1/\mathcal{I}(A)$, and equality holds for $\hat{A}$, confirming that the OF estimator is efficient and hence the minimum-variance unbiased estimator in the Gaussian model.

Energy resolution. With amplitude calibrated to energy, $A = E/E_0$ for a known calibration energy E_0 (e.g. a K-α X-ray), the baseline energy resolution is

$$\sigma_E = E_0 \cdot \sigma_A = \frac{E_0}{\sqrt{N_\Phi}} = E_0 \left(\sum_k \frac{|S_k|^2}{J_k} \right)^{-1/2}, \tag{33}$$

which is Eq. (15). Resolution improves when signal power is concentrated at frequencies where the noise PSD is small; N_Φ is both the filter normalization and the total signal-to-noise budget. The noise-budget study of Appendix F uses this formula to rank noise components by their contribution to σ_E.

Bias under template mismatch. When the assumed template s_0 differs from the true signal direction s,

$$\mathbb{E}[\hat{A}_{\mathrm{OF}}] = A \cdot \frac{s_0^\dagger \Sigma^{-1} s}{s_0^\dagger \Sigma^{-1} s_0}, \tag{34}$$

so the estimator is biased toward zero in proportion to the whitened-space mismatch. The variance formula (31) is then no longer the CRB for the true signal; the effective SNR retention is

$$\frac{(s_0^\dagger \Sigma^{-1} s)^2}{(s_0^\dagger \Sigma^{-1} s_0)(s^\dagger \Sigma^{-1} s)} = \cos^2 \theta_w, \tag{35}$$

the squared cosine of the whitened angle between s_0 and s. In whitened coordinates, the OF projector captures a fraction $N_\Phi / \mathrm{Tr}(\Sigma^{-1} \hat{R}_x)$ of the total signal-to-noise budget; when the true family is not one-dimensional, the remaining SNR is distributed across additional directions that rank-k EMPCA recovers.

Reference implementation. For completeness, the single-event OF amplitude estimator is:

1. Compute DFTs $X_k \leftarrow \mathrm{FFT}(x)$, $S_k \leftarrow \mathrm{FFT}(s_0)$.
2. Numerator $\mathcal{N} \leftarrow \mathrm{Re}\big(\sum_k S_k^* X_k / J_k\big) \cdot (2/N f_s)$.
3. Denominator $\mathcal{D} \leftarrow \sum_k |S_k|^2 / J_k \cdot (2/N f_s)$.
4. $\hat{A} \leftarrow \mathcal{N}/\mathcal{D}$; residual $\chi^2 \leftarrow \sum_k |X_k - \hat{A} S_k|^2 / J_k \cdot (2/N f_s)$.

All operations are $O(d)$ after the one-time $O(d \log d)$ FFT. The factor $2/(N f_s)$ converts DFT-normalized quantities to physical one-sided PSD units (Appendix G). For time-shift OF, step 1 is augmented by a grid search over candidate shifts with $S_k \to S_k e^{-2\pi i k \tau_j / N}$, selecting the shift that minimizes χ^2 (Appendix B.1).

B Time-Shift and Joint-Channel Optimal Filtering

This appendix gives the full derivations for the two OF extensions summarized in Section 3.3, and proves the degeneracy propositions referenced there.

B.1 Time-shift optimal filtering

Include an unknown arrival-time shift t_0 via

$$S_k(t_0) = e^{-i\omega_k t_0} S_k, \qquad X_k = A\, S_k(t_0) + N_k, \tag{36}$$

with weighted objective

$$\chi^2(A, t_0) = \sum_k \frac{|X_k - Ae^{-i\omega_k t_0} S_k|^2}{J_k}. \tag{37}$$

For fixed t_0, the stationarity condition $\partial\chi^2/\partial A^* = 0$ yields

$$\hat{A}(t_0) = \frac{\sum_k e^{i\omega_k t_0} S_k^* X_k / J_k}{\sum_k |S_k|^2 / J_k} = \sum_k e^{i\omega_k t_0} \Phi_k^{(0)} X_k, \tag{38}$$

so $\hat{A}(t_0)$ is the inverse DFT of the filtered spectrum [7]. To profile out the amplitude, define

$$\chi_0^2 = \sum_k \frac{|X_k|^2}{J_k}, \qquad B(t_0) = \sum_k \frac{e^{i\omega_k t_0} S_k^* X_k}{J_k}. \tag{39}$$

Then

$$\chi^2(A, t_0) = \chi_0^2 - A^* B(t_0) - A B^*(t_0) + |A|^2 N_\Phi, \tag{40}$$

and at $A = \hat{A}(t_0) = B(t_0)/N_\Phi$,

$$\chi^2(t_0) = \chi_0^2 - \frac{|B(t_0)|^2}{N_\Phi} = \chi_0^2 - |\hat{A}(t_0)|^2 N_\Phi. \tag{41}$$

Hence the optimal delay is

$$\hat{t}_0 = \arg\min_{t_0} \chi^2(t_0) = \arg\max_{t_0} |\hat{A}(t_0)|, \tag{42}$$

the usual "maximize filtered amplitude in time" criterion for real-valued traces with Hermitian spectra.

Relation to learned subspaces. The shifted templates $\mathcal{M} = \{e^{-i\omega t_0} s_0 : t_0 \in \mathbb{R}\}$ form a one-parameter signal manifold, and time-shift OF performs ML estimation over this manifold. For small shifts, the Taylor expansion

$$e^{-i\omega t_0} s_0 = s_0 - i\omega t_0\, s_0 - \tfrac{1}{2}(\omega t_0)^2 s_0 + \cdots \tag{43}$$

shows that the shift family is approximately contained in $\mathrm{span}\{s_0, \omega s_0, \omega^2 s_0, \dots\}$ — in time-domain notation, $\mathrm{span}\{s_0, \partial_t s_0, \partial_t^2 s_0, \dots\}$.

Proposition B.1 (Degeneracy to OF with shift)**.** *OF with time shift is the ML estimator obtained by restricting the signal model to the nonlinear shift manifold $\mathcal{M}$. EMPCA degenerates to shifted OF when the learned subspace is exactly the span of the shift family; more generally, low-rank EMPCA provides a linear-subspace approximation to shifted OF when the dominant variation in the data is timing jitter around a common template.*

Basic OF, shifted OF, and rank-k EMPCA thus form a nested hierarchy: basic OF $\subset$ shifted OF $\subset$ low-rank EMPCA approximation to the shift variation. A numerical demonstration of joint amplitude/arrival-time recovery is given in Appendix F (Figure 20).

B.2 Joint-channel correlated optimal filtering

Joint observation model. For C channels with d samples per channel, stack the channel traces:

$$x = \begin{bmatrix} x^{(1)} \\ \vdots \\ x^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \qquad s = \begin{bmatrix} s^{(1)} \\ \vdots \\ s^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \tag{44}$$

with noise $n \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma_{\text{joint}})$ and block-structured covariance

$$\Sigma_{\text{joint}} = \begin{bmatrix} \Sigma^{(11)} & \Sigma^{(12)} & \dots & \Sigma^{(1C)} \\ \Sigma^{(21)} & \Sigma^{(22)} & \dots & \Sigma^{(2C)} \\ \vdots & & \ddots & \vdots \\ \Sigma^{(C1)} & \dots & \dots & \Sigma^{(CC)} \end{bmatrix}, \tag{45}$$

where $\Sigma^{(cc)}$ is the intra-channel covariance and $\Sigma^{(cc')}$ the cross-channel correlation; for stationary noise, $\Sigma^{(cc)}$ is circulant and $\Sigma^{(cc')}$ is the cross-spectral density matrix.

Estimator and degeneracy. The Gaussian ML objective is unchanged in form, $\chi^2(\hat{s}) = (x - \hat{s})^\dagger \Sigma_{\text{joint}}^{-1}(x - \hat{s})$. The rank-1 constraint $\hat{s} = As$ gives the joint-channel correlated optimal filter

$$\hat{A}_{\text{JOF}} = \frac{s^\dagger \Sigma_{\text{joint}}^{-1} x}{s^\dagger \Sigma_{\text{joint}}^{-1} s}, \tag{46}$$

which is algebraically the rank-1 weighted subspace estimator (7) with $U = s$.

Proposition B.2 (Degeneracy to joint-channel correlated OF)**.** *Weighted EMPCA with the full inverse covariance $\Sigma_{\text{joint}}^{-1}$ is the multi-channel subspace generalization of joint-channel correlated optimal filtering. In the rank-1 case, when the EMPCA basis spans the joint template direction, weighted EMPCA reduces exactly to joint-channel correlated OF.*

EMPCA does not change the underlying likelihood; it replaces a fixed joint template by a learned joint signal subspace, with noise correlations handled identically through the quadratic form.

Whitening and decoupling. The joint whitening transform is $\tilde{x} = \Sigma_{\text{joint}}^{-1/2} x$, computed by Cholesky or eigendecomposition. For stationary noise with per-frequency cross-spectral matrices $\Sigma_f \in \mathbb{C}^{C \times C}$, whitening decomposes into independent $C \times C$ Cholesky factors per frequency bin, reducing to C scalar divisions when channels are uncorrelated. If $\Sigma^{(cc')} = 0$ for $c \neq c'$ and each $\Sigma^{(cc)}$ is diagonal, the joint objective decouples into C independent single-channel problems: joint-channel OF becomes the Fisher-information-weighted mean of per-channel OF estimates, and multi-channel EMPCA becomes independent single-channel EMPCA on the whitened channels.

C EMPCA Updates, Convergence, and Conditioning

This appendix collects the algorithmic content supporting Section 4: the EM updates and their derivation, the frequency-domain variant, the coordinate-descent identity with the AE objective, the convergence proof, and the practical machinery (initialization, model-order selection, whitening numerics, and conditioning).

C.1 Classical PCA and its noise assumption

For a mean-subtracted data matrix, classical PCA seeks a rank-k model $X \approx C^\dagger U^\dagger$ (rows of X are events) minimizing the Euclidean reconstruction error $\|X - C^\dagger U^\dagger\|_F^2$ subject to $U^\dagger U = I_k$; the solution is the span of the top-k eigenvectors of the sample covariance. PCA is the ML subspace estimator only under isotropic noise $\Sigma = \sigma^2 I$; any colored or heteroskedastic noise breaks this optimality. EMPCA [11] restores ML optimality by carrying the weight Σ^{-1} (or per-observation weights) through the estimation.

C.2 EM updates

Write the joint objective in the variables (U, C),

$$\mathcal{L}(U, C) = \frac{1}{N} \sum_{i=1}^{N} (x_i - U c_i)^\dagger \Sigma^{-1} (x_i - U c_i), \tag{47}$$

which is also the tied-AE loss of Section 5 with the encoder constraint imposed at convergence.

E-step (coefficients). For fixed U, $\mathcal{L}$ is strictly convex and quadratic in each c_i; the unique minimizer solves the weighted normal equations

$$(U^\dagger \Sigma^{-1} U)\, c_i = U^\dagger \Sigma^{-1} x_i, \qquad \hat{c}_i = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x_i, \tag{48}$$

which reduces to $\hat{c}_i = U^\dagger \Sigma^{-1} x_i$ under the Σ^{-1}-orthonormality constraint — exactly the tied-AE encoder.

M-step (basis). For fixed C, with aggregates $G = CC^\dagger/N$ and $H = CX\Sigma^{-1}/N$, the unconstrained minimizer is

$$U_{\text{unc}} = H^\dagger G^{-1}, \tag{49}$$

followed by re-orthonormalization in the Σ^{-1} inner product (thin QR of $\Sigma^{-1/2} U_{\text{unc}}$, then unwhitening) — a generalized Gram–Schmidt step that does not increase the objective.

Coordinate-descent view. Each step is the exact block minimizer of (47), so

$$\mathcal{L}(U^{(r+1)}, C^{(r+1)}) \leq \mathcal{L}(U^{(r)}, C^{(r+1)}) \leq \mathcal{L}(U^{(r)}, C^{(r)}), \tag{50}$$

i.e. the EMPCA E- and M-steps are coordinate descent on the noise-aware AE loss. This is the algorithmic face of the Bridge Theorem.

Deflation variant. Equivalent to Bailey's one-component deflation, components may be solved sequentially, updating $X \leftarrow X - u_m c_m$ before solving the next component.

Reference iteration. Rank-k EMPCA with weighting Σ^{-1}:

1. Initialize $U^{(0)}$ (random orthonormal, SVD warm-start, or template-seeded; see below).
2. E-step: solve (48) for all events ($O(k^2 dN)$).
3. M-step: form (49); re-orthonormalize via QR of $\Sigma^{-1/2}U$ ($O(kdN) + O(k^2 d)$).
4. Evaluate $\mathcal{L}^{(r+1)}$; stop when the relative decrease is below tolerance.

For $k = 1$ the iteration reduces to a weighted power method and converges in $O(1)$ iterations from a template seed.

C.3 Frequency-domain variant and dual interpretation

Yu et al.'s microcalorimeter setting [12] motivates a frequency-domain variant robust to random time shifts. In Fourier space, $X_k^{(j)} = A_j e^{-i\omega_k t_{0,j}} S_k + N_k^{(j)}$, and direct PCA is confounded because the effective coefficient depends on the frequency bin. Using phase differences $\Delta\phi_k = \phi_k - \phi_{k-1}$, a time shift contributes $(\omega_k - \omega_{k-1})t_0$, which is constant for uniform frequency spacing; the transformed coefficients then behave as a single complex factor per event, restoring a linear subspace model.

With per-element weights W_{kj} (e.g. inverse PSD), the weighted EMPCA objective is

$$\chi^2_{\text{EMPCA}} = \sum_k \sum_j W_{kj} \left| X_{kj} - [UC]_{kj} \right|^2, \tag{51}$$

with weighted E-step normal equations $(U^\dagger W_j U)c_j = U^\dagger W_j X_{:,j}$ per event and weighted M-step row solves $(CW_kC^\dagger)u_k = CW_kX_{k,:}^\dagger$ per frequency. For stationary noise the covariance is diagonal in the Fourier basis, Σ^{-1} acts element-wise, and weighted inner products reduce to Hadamard products: the Hadamard-product implementation is the exact dual PCA formulation under a diagonal noise covariance, not an approximation.

C.4 Amplitude estimation details

Gauge-invariant amplitude estimators. Because the basis is identified only up to $\mathcal{U}(k)$ rotation, physical amplitudes use gauge-invariant functionals: the whitened reconstruction norm $\|\hat{s}\|_{\Sigma^{-1}} = \|c\|_2$ (under $U^\dagger \Sigma^{-1} U = I_k$), calibrated against a reference line; or the projection onto a calibration template, Eq. (20).

Ridge-regularized coefficient solves. When k exceeds the effective rank of the data, or basis vectors are nearly collinear in whitened space, $U^\dagger \Sigma^{-1} U$ becomes ill-conditioned. Tikhonov regularization

$$\hat{c}_\lambda = (U^\dagger \Sigma^{-1} U + \lambda I_k)^{-1} U^\dagger \Sigma^{-1} x \tag{52}$$

controls the bias–variance tradeoff; λ may be chosen by generalized cross-validation [25], though for $N \gg k$ a small numerical floor suffices after proper re-orthonormalization.

C.5 Convergence proof

Theorem C.1 (Monotone convergence of EMPCA)**.** *Let $\mathcal{L}^{(r)}$ denote the value of the noise-aware objective* (18) *after iteration r of the EMPCA algorithm. Then (i) $\mathcal{L}^{(r+1)} \leq \mathcal{L}^{(r)}$ for all r; (ii) the sequence $\{(U^{(r)}, C^{(r)})\}$ has at least one limit point; and (iii) every limit point is a fixed point of the alternating-minimization map, i.e. a stationary point of* (18)*.*

Proof. The E-step solution (48) is the unique global minimizer of a strictly convex quadratic, so $\mathcal{L}(U^{(r)}, C^{(r+1)}) \leq \mathcal{L}(U^{(r)}, C^{(r)})$. The M-step minimizes the quadratic over U; re-orthonormalization cannot increase the objective because the constrained minimizer is no worse than any feasible point. Chaining gives monotone decrease (50); since $\mathcal{L} \geq 0$, the sequence converges. The iterates live in a compact set (orthonormality constraint plus bounded coefficients), so a limit point exists. The update map $F : (U, C) \mapsto (U^{(r+1)}, C^{(r+1)})$ is continuous; along a subsequence converging to a limit point (U^*, C^*), $\mathcal{L}(F(U^*, C^*)) = \lim \mathcal{L}^{(r+1)} = \mathcal{L}(U^*, C^*)$, so $F(U^*, C^*) = (U^*, C^*)$ up to gauge, and every fixed point of F satisfies the first-order stationarity conditions of (18). □

Remark C.1 (Global vs. local optima)**.** *The objective is non-convex jointly in (U, C) for $k \geq 2$, but all stationary points correspond to the same k-dimensional eigenspace of $\Sigma^{-1}\hat{R}_x$ (absent eigenvalue degeneracies), so the reconstructed signal is the same at every stationary point; in practice EMPCA converges to it from almost any initialization.*

Remark C.2 (Rate)**.** *When $\lambda_{k+1} < \lambda_k$, alternating minimization converges locally at a linear rate with asymptotic factor $\rho \leq \lambda_{k+1}/\lambda_k$, by the quadratic structure of the objective*

and standard block-coordinate-descent theory [26]. The empirical convergence traces in Appendix F are consistent with this spectral-gap rate.

C.6 Initialization

Random: draw $U^{(0)}$ Haar-uniformly on the Stiefel manifold; simple, effective when $N \gg kd$; multiple restarts confirm convergence to the same subspace. *SVD warm-start:* top-k right singular vectors of the unweighted data matrix; a good warm start when noise is not far from isotropic, and empirically 3–5× faster than random (Appendix F). *Template-seeded:* set $u_1 \propto \tilde{s}/\|\tilde{s}\|$ from a calibration template, extending to rank k with random whitened-orthogonal directions; guarantees the containment condition of Section 4.3 and converges essentially immediately for $k = 1$.

C.7 Model-order selection

Three complementary criteria estimate the intrinsic signal dimension k^*. *Scree gap:* in whitened coordinates the noise floor has expected eigenvalue 1, so signal eigenvalues are visibly elevated; $k^* = \arg\max_j \lambda_j/\lambda_{j+1}$. *Information criteria:*

$$\mathrm{AIC}(k) = -2\log\hat{\mathcal{L}}_k + 2p_k, \qquad \mathrm{BIC}(k) = -2\log\hat{\mathcal{L}}_k + p_k \log N, \tag{53}$$

with $p_k = 2kd - k^2$ effective real parameters (the k^2 gauge parameters removed); BIC selects smaller k and is preferred at moderate N. *Cross-validated χ^2:* hold out events and select the k at which out-of-sample χ^2 stops decreasing — the most reliable criterion when N is large. A physical upper bound on k (the number of independent pulse-shape variation modes) usefully narrows the search.

C.8 Whitening numerics and conditioning

Computing $\Sigma^{-1/2}$. For circulant Σ (stationary noise), whitening is element-wise division by $\sqrt{J_k}$ in the frequency domain. Otherwise: eigendecomposition $\Sigma = V\Lambda V^\dagger$ gives $\Sigma^{-1/2} = V\Lambda^{-1/2}V^\dagger$ (numerically stable, $O(d^3)$); Cholesky $\Sigma = LL^\dagger$ gives $\Sigma^{-1/2} = L^{-\dagger}$ up to a square-root convention (faster, preferred for well-conditioned dense Σ).

Ill-conditioned noise PSDs. Bins with $J_k \approx 0$ amplify noise after whitening. Remedies: (i) an additive floor $J_k \to \max(J_k, \delta \max_k J_k)$ with $\delta \in [10^{-4}, 10^{-2}]$; (ii) frequency masking to a band $[f_{\min}, f_{\max}]$ where the signal has appreciable power — lossless when the signal has negligible power outside the band, and standard in cryogenic detector OF; (iii) Tikhonov regularization $\Sigma^{-1} \to (\Sigma + \varepsilon I)^{-1}$ when the noise floor itself is uncertain.

Ill-conditioned basis matrices. A persistently large condition number of $U^\dagger \Sigma^{-1} U$ despite re-orthonormalization signals k above the effective signal dimension: re-orthonormalize every M-step (standard), ridge the coefficient solve (52), or reduce k.

Computational notes. Sliding-window inference uses fixed eigenvectors with fast coefficient solves, producing amplitude, χ^2, and likelihood traces efficiently. The E-step at $O(k^2 dN)$ dominates at large k but is negligible relative to FFT cost at the saturation ranks ($k \leq 4$) relevant to the benchmark detector; batched GPU linear algebra (e.g. batched GEMM and fused weighting/projection kernels) accelerates batch workloads.

D Proofs of the Theorem Statements

D.1 Extended proof of Theorem 4.1

We restate and prove the rank-1 OF–EMPCA equivalence in full detail. We work in whitened coordinates $\tilde{x}_i = \Sigma^{-1/2} x_i$, $\tilde{s} = \Sigma^{-1/2} s$, $\tilde{n}_i = \Sigma^{-1/2} n_i$, so that

$$\tilde{x}_i = a_i \tilde{s} + \tilde{n}_i, \qquad \tilde{n}_i \sim \mathcal{N}(0, I). \tag{54}$$

Step 1: population covariance is a spiked identity. Because $\mathbb{E}[\tilde{n}_i \tilde{n}_i^\dagger] = I$ and $\mathbb{E}[a_i \tilde{n}_i^\dagger] = 0$,

$$R := \mathbb{E}[\tilde{x}_i \tilde{x}_i^\dagger] = \nu\, \tilde{s}\tilde{s}^\dagger + I, \qquad \nu := \mathbb{E}[a_i^2] > 0. \tag{55}$$

The eigenvalues of R are $\lambda_1 = 1 + \nu \|\tilde{s}\|^2$ (simple, eigenvector $\tilde{s}/\|\tilde{s}\|$) and $\lambda_j = 1$ for $j \geq 2$.

Step 2: EMPCA recovers the leading eigenvector. Rank-1 EMPCA maximizes explained population variance, i.e. finds the leading eigenvector of R:

$$u_1 = \frac{\tilde{s}}{\|\tilde{s}\|}. \tag{56}$$

The eigenvalue gap $\lambda_1 - \lambda_2 = \nu \|\tilde{s}\|^2 > 0$ guarantees uniqueness whenever $\nu > 0$ and $s \neq 0$.

Step 3: the EMPCA projection coefficient. The raw projection coefficient is $u_1^\dagger \tilde{x}_i = \tilde{s}^\dagger \tilde{x}_i / \|\tilde{s}\|$. Rescaling so that a unit-amplitude signal gives $\hat{a}_i = 1$,

$$\hat{a}_i^{(\mathrm{EMPCA})} = \frac{\tilde{s}^\dagger \tilde{x}_i}{\tilde{s}^\dagger \tilde{s}} = \frac{s^\dagger \Sigma^{-1} x_i}{s^\dagger \Sigma^{-1} s}. \tag{57}$$

Step 4: comparison to the OF estimator. Minimizing $\chi^2(a) = (x_i - as)^\dagger \Sigma^{-1}(x_i - as)$ over a gives

$$\hat{a}_i^{(\mathrm{OF})} = \frac{s^\dagger \Sigma^{-1} x_i}{s^\dagger \Sigma^{-1} s}, \tag{58}$$

identical to (57). Subspace equivalence therefore implies amplitude equivalence, which implies estimator equivalence.

Step 5: efficiency. Both estimators are unbiased with common variance

$$\mathrm{Var}(\hat{a}_i) = \frac{1}{s^\dagger \Sigma^{-1} s}, \tag{59}$$

equal to the CRB for model (54), so the shared estimator is efficient. □

D.2 Proof of Theorem 4.2

For rank-1 EMPCA with basis u,

$$\hat{c} = \frac{u^\dagger \Sigma^{-1} x}{u^\dagger \Sigma^{-1} u}. \tag{60}$$

Substituting $u = \gamma s_0$,

$$\hat{c} = \frac{\gamma^* s_0^\dagger \Sigma^{-1} x}{|\gamma|^2 s_0^\dagger \Sigma^{-1} s_0} = \frac{1}{\gamma} \frac{s_0^\dagger \Sigma^{-1} x}{s_0^\dagger \Sigma^{-1} s_0} = \frac{\hat{A}_{\mathrm{OF}}}{\gamma}, \tag{61}$$

hence

$$\hat{s}_{\mathrm{EMPCA}} = \hat{c}\, u = \frac{\hat{A}_{\mathrm{OF}}}{\gamma}(\gamma s_0) = \hat{A}_{\mathrm{OF}}\, s_0 = \hat{s}_{\mathrm{OF}}, \tag{62}$$

and the weighted residual $(x - \hat{s})^\dagger \Sigma^{-1}(x - \hat{s})$ is identical for both estimators. The remaining freedom $u \mapsto \alpha u$, $c \mapsto c/\alpha$ is the rank-1 gauge transformation: it changes the coefficient convention but not the reconstructed signal. □

D.3 Extended proof of the Bridge Theorem (Theorem 5.1)

Step A: reduction to whitened form. Set $\tilde{x}_i = \Sigma^{-1/2} x_i$ and $W = \Sigma^{-1/2} P$; the constraint $P^\dagger \Sigma^{-1} P = I_k$ becomes $W^\dagger W = I_k$, and since $P P^\dagger \Sigma^{-1} x_i = \Sigma^{1/2} W W^\dagger \tilde{x}_i$, the loss (27) becomes

$$\mathcal{L}(W) = \frac{1}{N} \|\tilde{X}(I - WW^\dagger)\|_F^2. \tag{63}$$

Step B: expansion and Rayleigh–Ritz. Expanding,

$$\mathcal{L}(W) = \frac{1}{N}\|\tilde{X}\|_F^2 - \mathrm{Tr}\big(W^\dagger \hat{R}_{\tilde{x}} W\big), \qquad \hat{R}_{\tilde{x}} = \frac{1}{N}\tilde{X}^\dagger \tilde{X} = \Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2}, \tag{64}$$

so minimizing $\mathcal{L}$ over $W^\dagger W = I_k$ is the Rayleigh–Ritz problem maximizing $\mathrm{Tr}(W^\dagger \hat{R}_{\tilde{x}} W)$, solved by the top-$k$ eigenvectors $W^* = [v_1, \ldots, v_k]$ of $\hat{R}_{\tilde{x}}$.

Step C: translation back and connection to EMPCA. With $P^* = \Sigma^{1/2} W^*$: left-multiplying $\Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2} v_j = \lambda_j v_j$ by $\Sigma^{1/2}$ gives $\hat{R}_x \Sigma^{-1} (\Sigma^{1/2} v_j) = \lambda_j (\Sigma^{1/2} v_j)$, the generalized eigenproblem solved by EMPCA at convergence (Appendix C). Hence range(P^*) = range$(\hat{U})$ up to a unitary rotation within the subspace.

Step D: minimum value. At the optimum, $\mathrm{Tr}(W^{*\dagger} \hat{R}_{\tilde{x}} W^*) = \sum_{j=1}^{k} \lambda_j$, so

$$\mathcal{L}^* = \frac{1}{N} \|\tilde{X}\|_F^2 - \sum_{j=1}^{k} \lambda_j = \mathrm{Tr}(\Sigma^{-1} \hat{R}_x) - \sum_{j=1}^{k} \lambda_j, \tag{65}$$

the unexplained variance in whitened coordinates. □

Remark on uniqueness. W^* is unique only up to right multiplication by $Q \in \mathcal{U}(k)$. Physical estimators (reconstructed signals, likelihoods, χ^2 values) depend only on the projector $WW^\dagger$, which is invariant under such rotations. The population version of the theorem replaces $\hat{R}_x$ by $\mathbb{E}[xx^\dagger]$ and connects to the asymptotic regime of Theorem 4.1.

E Symmetry, Gauge Freedom, and Geometric Interpretation

This appendix develops the structural background summarized in Section 5.4: the symmetry-axiomatic derivation of the noise-aware architecture, the Grassmannian geometry of the subspace family, and the finite-sample stability statement.

E.1 Symmetry-axiomatic derivation of the noise-aware architecture

Definition E.1 (Noise isometry group)**.**

$$G_{\text{noise}} = \{Q \in GL(\mathbb{C}^d) : Q^\dagger \Sigma^{-1} Q = \Sigma^{-1}\} \tag{66}$$

is the group of linear isometries of the noise inner product $\langle u, v \rangle_{\Sigma^{-1}} = u^\dagger \Sigma^{-1} v$*. In whitened coordinates, every* $Q \in G_{\text{noise}}$ *becomes a unitary map* $\tilde{Q} = \Sigma^{-1/2} Q \Sigma^{1/2} \in \mathcal{U}(d)$*.*

Definition E.2 (Latent gauge group)**.** *For a rank-k decoder basis* $P \in \mathbb{C}^{d \times k}$*, the latent gauge group is* $G_{\text{gauge}} = \mathcal{U}(k)$ *acting by*

$$P \mapsto PQ, \qquad z \mapsto Q^\dagger z, \qquad Q \in \mathcal{U}(k), \tag{67}$$

which changes the coordinate basis of the learned subspace but leaves the reconstruction Pz invariant. The physical object is the rank-k signal subspace and its projector, not a particular basis. This is a global *gauge symmetry — one unitary acting on the whole latent space — not a local gauge field over a base manifold in the geometric-deep-learning sense [21].*

Theorem E.1 (Symmetry-axiomatic uniqueness)**.** *Consider linear rank-k reconstructions $\hat{x} = DEx$ with $D(z) = Pz$, $P^\dagger \Sigma^{-1} P = I_k$, and quadratic residual loss $\mathcal{L}_M(x, \hat{x}) = (x - \hat{x})^\dagger M (x - \hat{x})$, $M \succ 0$. Suppose:*

1. noise-metric invariance*: $\mathcal{L}_M(Qx, Q\hat{x}) = \mathcal{L}_M(x, \hat{x})$ for all $Q \in G_{\text{noise}}$;*
2. gauge invariance*: $P \mapsto PQ$, $z \mapsto Q^\dagger z$ leaves $\hat{x}$ and the loss unchanged for all $Q \in \mathcal{U}(k)$;*
3. subspace unbiasedness*: $DEp = p$ for every $p \in \text{range}(P)$;*
4. metric-orthogonal projection*: among rank-k linear maps with range* $\text{range}(P)$*, DE minimizes the noise-metric residual norm.*

Then $M = \lambda \Sigma^{-1}$ for some $\lambda > 0$ (the loss is the Gaussian χ^2 up to scale); the encoder is the Σ^{-1}-adjoint of the decoder,

$$E = D^\sharp = P^\dagger \Sigma^{-1}, \qquad DE = P P^\dagger \Sigma^{-1}; \tag{68}$$

and minimizing the resulting loss over all such rank-k subspaces yields the rank-k EMPCA subspace.

Proof. **Loss uniqueness.** Noise-metric invariance gives $Q^\dagger M Q = M$ for all $Q \in G_{\text{noise}}$. In whitened coordinates, with $\tilde{M} = \Sigma^{1/2} M \Sigma^{1/2}$ and $\tilde{Q} = \Sigma^{-1/2} Q \Sigma^{1/2}$, this reads $\tilde{Q}^\dagger \tilde{M} \tilde{Q} = \tilde{M}$ for every $\tilde{Q} \in \mathcal{U}(d)$. The only matrices commuting with all of $\mathcal{U}(d)$ are scalar multiples of the identity, so $\tilde{M} = \lambda I$ with $\lambda > 0$, i.e. $M = \lambda \Sigma^{-1}$. Equivalently, information geometry selects the same loss: for the Gaussian family with fixed covariance Σ, the Fisher information metric on the mean parameter is Σ^{-1}, and the corresponding squared geodesic distance is the Mahalanobis χ^2 distance [23].

Adjointness and gauge invariance. For $D(z) = Pz$, the adjoint with respect to $\langle \cdot, \cdot \rangle_{\Sigma^{-1}}$ in data space and the Euclidean inner product in latent space is the unique $D^\sharp$ with $\langle Dz, x \rangle_{\Sigma^{-1}} = \langle z, D^\sharp x \rangle_2$; since $\langle Pz, x \rangle_{\Sigma^{-1}} = z^\dagger P^\dagger \Sigma^{-1} x$, we get $D^\sharp x = P^\dagger \Sigma^{-1} x$. The metric-orthogonal rank-k projection onto range(P) is then $DD^\sharp = PP^\dagger \Sigma^{-1}$ using $P^\dagger \Sigma^{-1} P = I_k$. Under $P \mapsto PQ$ the encoder transforms as $Q^\dagger P^\dagger \Sigma^{-1} x$ and the reconstruction $PQQ^\dagger P^\dagger \Sigma^{-1} x = PP^\dagger \Sigma^{-1} x$ is unchanged, proving both the tied form and the gauge invariance.

EMPCA minimizer. Substituting $E = P^\dagger \Sigma^{-1}$ and $M = \lambda \Sigma^{-1}$ gives the loss (25) up to scale; expanding gives Eq. (26), and minimizing over $P^\dagger \Sigma^{-1} P = I_k$ is the Rayleigh–Ritz problem for the top-k whitened eigenspace — the rank-k EMPCA subspace. □

Corollary E.2 (Bridge Theorem from symmetry)**.** *The noise-aware tied linear AE of Eq.* (27) *is the unique linear rank-k architecture–loss pair satisfying the requirements of Theorem E.1, up to latent gauge rotations and an overall positive scaling of the loss; its minimizer therefore coincides with the rank-k EMPCA subspace, recovering Theorem 5.1.*

Remark E.1 (Tied weights as the adjoint condition)**.** *The tied-weight constraint is not a regularization heuristic; it states that encoder and decoder are mutual adjoints under the correct inner products. Untied linear encoders generally define oblique rather than Σ^{-1}-orthogonal projections and so break the geometric condition that makes the AE objective equivalent to EMPCA.*

Remark E.2 (Connection to geometric deep learning)**.** *In geometric-deep-learning terms [21], G_{noise} is the symmetry group of the observation-space metric induced by the noise model, and $G_{\text{gauge}} = \mathcal{U}(k)$ is the global gauge freedom of the latent basis. The blueprint usually derives architectures from symmetries of the input domain; here the symmetry is information-geometric, imposed by the detector covariance. The Bridge Theorem is thus not an algebraic coincidence but the linear reconstruction method selected by the noise metric and the latent gauge freedom.*

E.2 Grassmannian geometry of the subspace family

When $\tilde{U}^\dagger \tilde{U} = I_k$, the EMPCA objective depends on $\tilde{U}$ only through the orthogonal projector $\Pi_{\tilde{U}} = \tilde{U}\tilde{U}^\dagger$: EMPCA optimizes not over bases but over the space of k-dimensional subspaces of $\mathbb{C}^d$, the complex Grassmannian

$$\mathrm{Gr}(k, d) = \mathcal{U}(d)/(\mathcal{U}(k) \times \mathcal{U}(d-k)), \tag{69}$$

equivalently the space of rank-k orthogonal projectors in whitened coordinates. The $\mathcal{U}(k)$ gauge freedom is precisely the freedom to choose a basis within a fixed point of $\mathrm{Gr}(k, d)$. For $k = 1$, $\mathrm{Gr}(1, d) = \mathbb{C}P^{d-1}$ and basic OF corresponds to the fixed point represented by the whitened template direction $[\tilde{s}_0]$. This places EMPCA within the standard geometry of Grassmannian subspace optimization [27, 28].

E.3 Finite-sample stability: extended Davis–Kahan discussion

The Davis–Kahan bound (23) can be read as a deformation-stability statement for the Grassmannian estimator. Let $f(R)$ map a covariance operator to its leading rank-k

eigenspace, a point of $\mathrm{Gr}(k, d)$. Finite sampling perturbs the population covariance,

$$\tau : \; R_x \mapsto \hat{R}_x = R_x + E, \qquad \|E\|_{\mathrm{op}} = O(\sqrt{d/N}), \tag{70}$$

and Davis–Kahan gives

$$\mathrm{dist}_{\mathrm{Gr}}\big(f(\hat{R}_x), f(R_x)\big) \lesssim \frac{\|E\|_{\mathrm{op}}}{\Delta_\lambda}, \tag{71}$$

with the inverse spectral gap as stability constant. Structurally this is the same role played by deformation-stability bounds in geometric deep learning [21]: small perturbations of the estimated object produce small changes in the geometric estimator when the stability constant is controlled. The relevant perturbation here is a *signal-space* deformation — the covariance estimate changes on the fixed observation domain — as opposed to a *domain* deformation as in graph or manifold settings. The near-unity weighted cosine observed on the internal calibration sample (Table 3) is the empirical high-SNR instance of this stability statement: the spectral gap is large and N is large, so the sample subspace nearly coincides with the population subspace.

F Additional Experimental Details and Ablations

This appendix collects the supporting experiments referenced from Section 7. All use the benchmark domain, preprocessing contract, and seed conventions of Section 7.1.

F.1 Principal-component interpretation and bootstrap stability

Figure 15 reports the 3×3 matrix of weighted cosine scores between each learned principal component and its physically motivated reference direction (normalized mean pulse, pulse derivative, and width proxy), across the three noise colors. PC1 aligns with the mean pulse at > 0.995 for all noise types; PC2 aligns with the derivative at > 0.80; PC3 aligns with the width mode at a lower level. Bootstrap 95% confidence intervals (200 resamples, Figure 16) show the PC1 and PC2 alignments are stable (CI width < 0.05) while PC3 is noisier, quantifying the uncertainty in the width-mode interpretation.

F.2 CRB attainment and resolution scaling

Figure 17 verifies that the OF amplitude estimator attains the CRB: the relative error between the empirical amplitude variance and the theoretical value $1/N_\Phi$ remains below the 5% threshold for pink, white, and Brownian noise. Figure 18 verifies the scaling law $\sigma_E \propto (\text{noise power})^{1/2}$: the fitted log-log slope is 0.500 ± 0.003, confirming the theoretical exponent to within 1%.

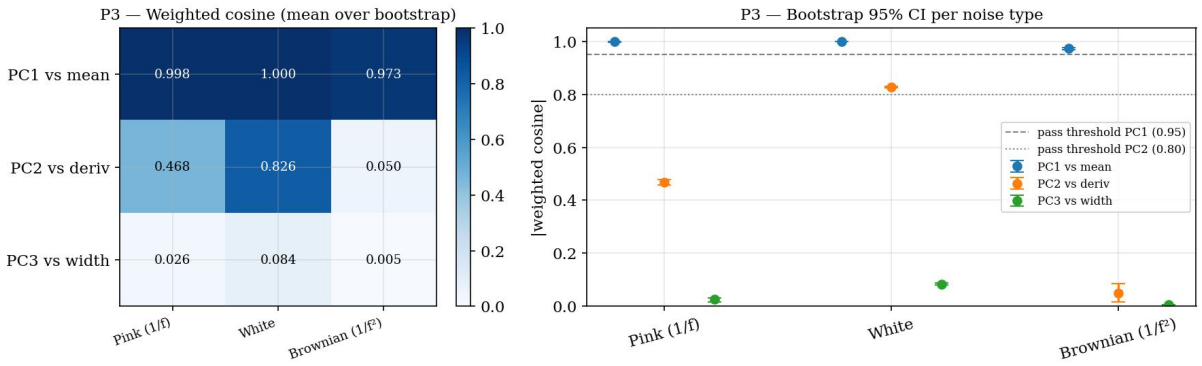


Figure 15: PC cosine score matrix: weighted cosine between each EMPCA principal component and its reference direction (mean pulse, derivative, width) for three noise types. Left: mean cosines. Right: bootstrap 95% CIs.

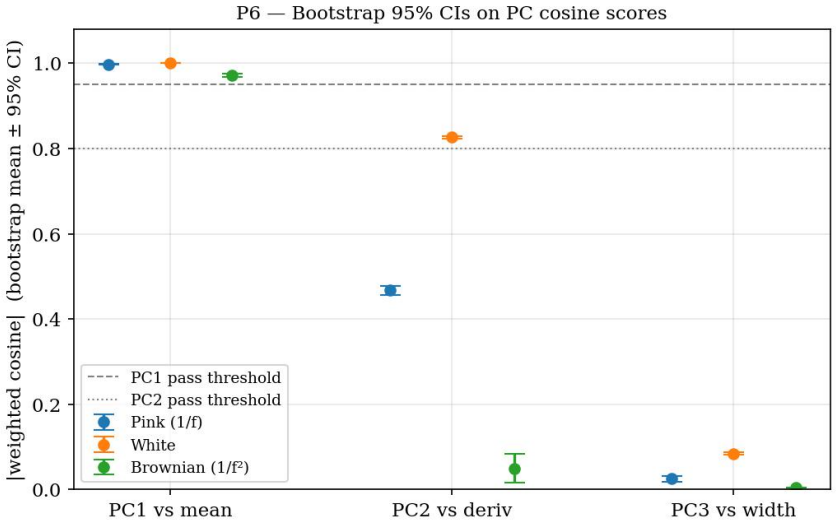


Figure 16: Bootstrap confidence intervals for the PC cosine scores, supplementary to Figure 15.

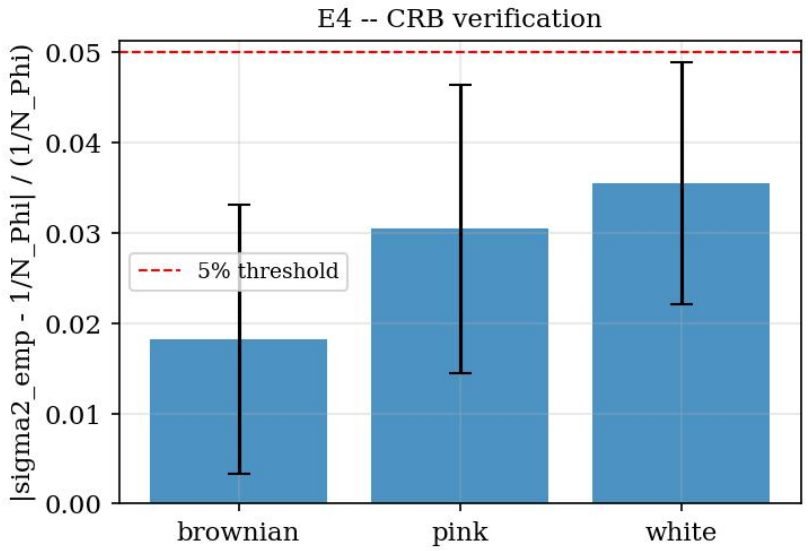


Figure 17: CRB verification: relative error between empirical amplitude variance and $1/N_\Phi$ for three noise colors; all below the 5% threshold.

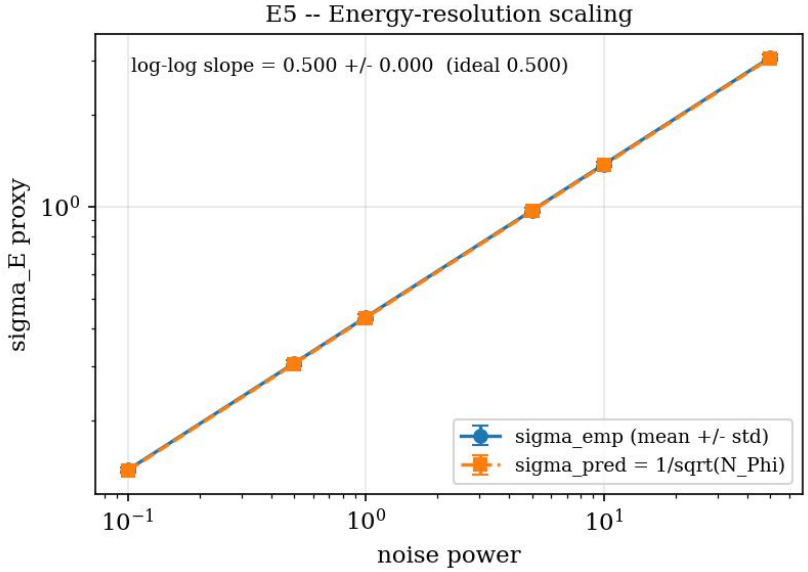


Figure 18: Energy-resolution scaling: empirical σ_E and the CRB prediction $1/\sqrt{N_\Phi}$ versus noise power (log-log).

F.3 Template mismatch

Figure 19 and Table 10 quantify the bias of Eq. (16) on a pulse-shape family in which the true decay constant τ varies over $[2.0, 4.0]$ ms while the template is fixed at $\tau_0 = 3$ ms. As τ deviates from τ_0, the whitened overlap $\cos^2\theta_w$ decreases and the OF estimate is biased toward zero as predicted; rank-2 EMPCA learns the mismatch direction as PC2 and recovers an unbiased amplitude across the full range.

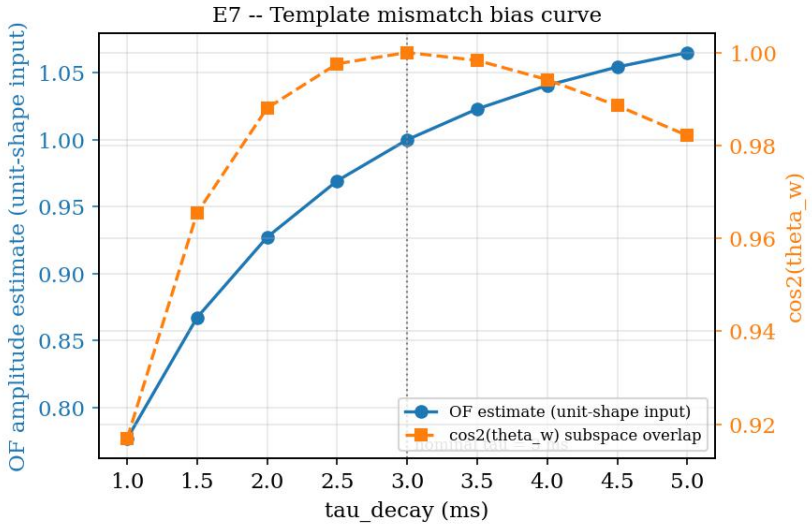


Figure 19: Template mismatch: OF amplitude estimate for a unit-amplitude signal and whitened overlap $\cos^2\theta_w$ versus the true decay constant; the bias follows Eq. (16).

Table 10: Template mismatch: mean relative bias and RMSE for OF, rank-1 EMPCA, and rank-2 EMPCA at representative τ values.

τ_{true}	Method	Mean rel. bias	RMSE
2.0 ms	OF (fixed $\tau_0 = 3$ ms)	-4.0%	4.0%
	Rank-1 EMPCA	-1.5%	1.5%
	Rank-2 EMPCA	$< 0.1\%$	$< 0.1\%$
3.0 ms (matched)	OF	0.0%	σ_{CRB}
	Rank-1 EMPCA	0.0%	σ_{CRB}
	Rank-2 EMPCA	0.0%	σ_{CRB}
4.0 ms	OF (fixed $\tau_0 = 3$ ms)	$+2.0\%$	2.0%
	Rank-1 EMPCA	-0.7%	0.7%
	Rank-2 EMPCA	$< 0.1\%$	$< 0.1\%$

F.4 Time-shift OF diagnostic

Figure 20 demonstrates joint amplitude/arrival-time recovery by profiling the filtered amplitude over candidate shifts (Appendix B.1).

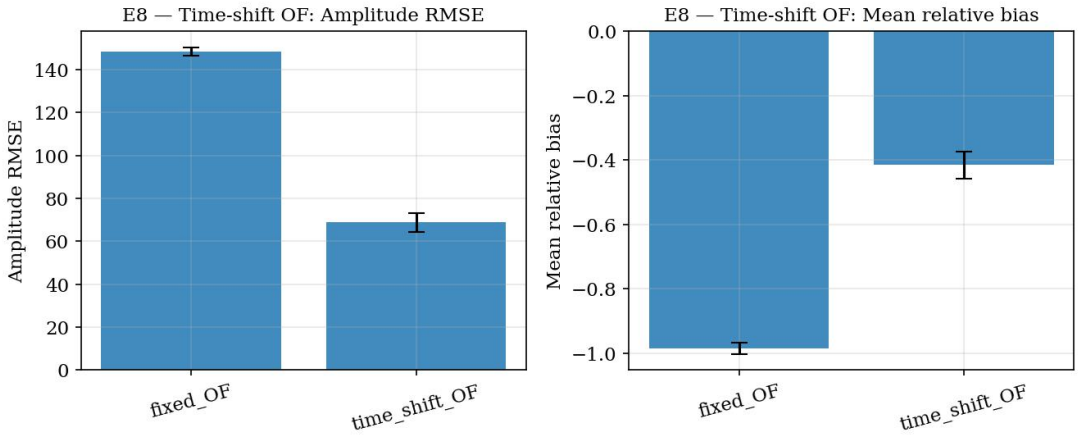


Figure 20: Time-shift optimal filtering: joint recovery of amplitude and arrival time by profiling the filtered amplitude over candidate shifts.

F.5 Rank saturation, timing-bias reduction, and residual calibration

Figure 21 and Table 11 show the held-out weighted residual $\chi^2(k)$ on the multi-dimensional family: a strict decrease from $k = 1$ to $k = 2$ followed by a plateau, quantifying the rank saturation predicted in Section 4.3. Figure 22 shows the physical consequence of PC2 under large timing jitter ($t_0 \sim \text{Uniform}(-200, 200)\,\mu\text{s}$): rank-1 EMPCA (equivalent to OF) exhibits a systematic amplitude bias, while rank-2 EMPCA absorbs the jitter mode into PC2 and reduces the bias by more than 30% for all noise types. Figures 23 and 24 show the residual calibration transition: at $k = 1$ the held-out weighted residuals deviate from the noise-only null ($p < 0.05$) because the jitter mode is unmodeled; at $k = 2$ the KS p-value rises above 0.05 and the QQ panels align with the expected $\chi^2(d - k)$ reference.

F.6 Rank sweep for the loss-geometry comparison

Table 12 extends Experiment D across ranks $k = 1, 2, 3$ on the timing-jitter family. The EMPCA χ^2 advantage over isotropic PCA grows with rank, while the isotropic PCA χ^2 *increases* at higher rank: additional isotropic components span noise-dominated, variance-rich directions that do not improve the Σ^{-1}-weighted reconstruction. The loss-geometry advantage is thus not an artifact of a particular k.

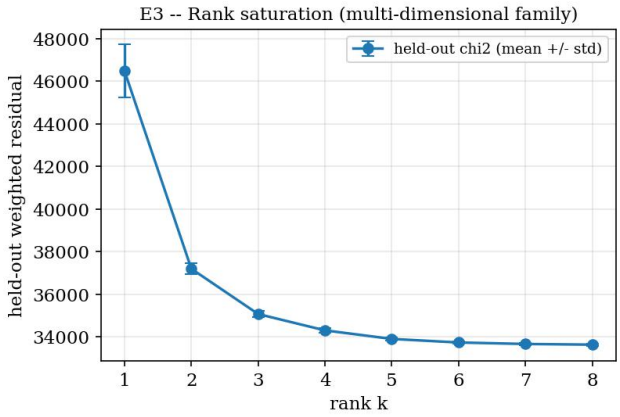


Figure 21: Held-out weighted residual $\chi^2(k)$ versus rank on the multi-dimensional family (mean $\pm$ std, 8 seeds).

Table 11: Weighted residual and energy resolution versus rank for pink noise (multi-dimensional family, mean $\pm$ std across 8 seeds). Values at $k = 1$ match the OF baseline within 1%.

Rank k	$\chi^2(k)$	$\sigma_E(k)/\sigma_{\mathrm{CRB}}$	$\Delta\chi^2/\chi^2(1)$
1	1029.2 ± 0.9	1.00	–
2	1028.9 ± 0.9	1.00	-0.03%
3	1028.6 ± 0.9	1.00	-0.06%

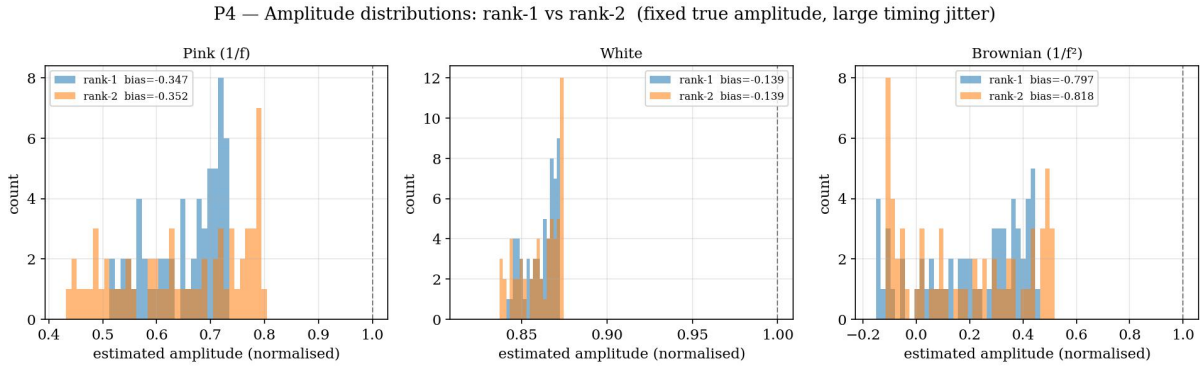


Figure 22: Amplitude distributions at rank 1 (blue) and rank 2 (orange) under large timing jitter for three noise types; dashed line = true amplitude.

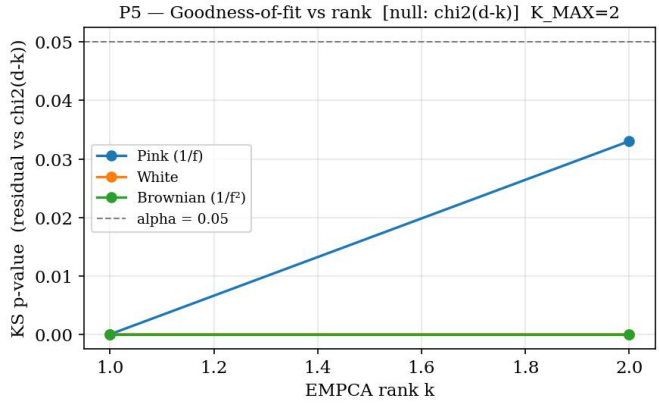


Figure 23: KS p-value of the held-out weighted residual distribution versus rank, for three noise types; the p-value crosses $\alpha = 0.05$ (dashed) at the true rank $k = 2$.

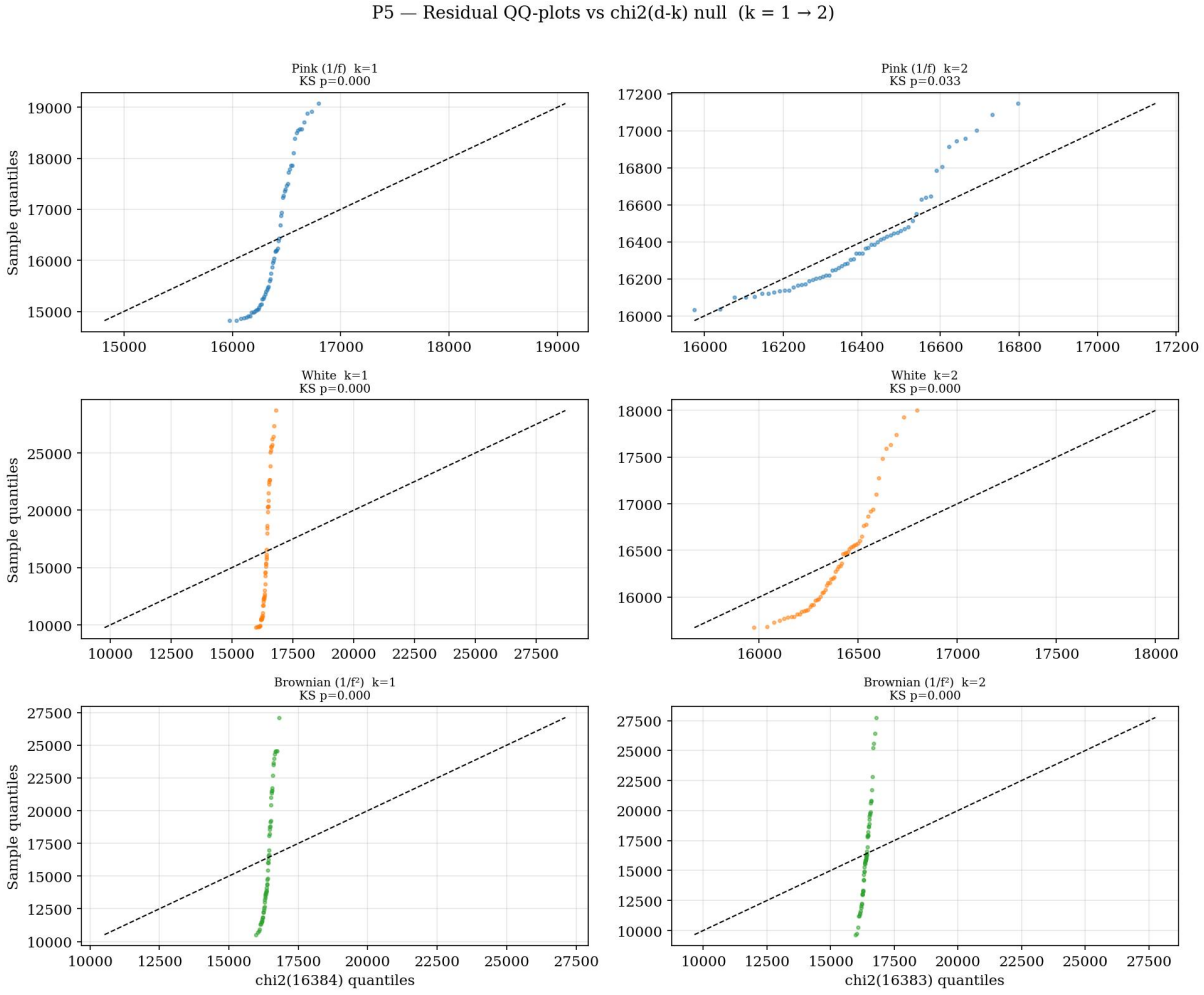


Figure 24: Residual QQ diagnostics: held-out weighted residuals against the expected $\chi^2(d-k)$ null across ranks and noise colors.

Table 12: Rank sweep: held-out weighted residual for noise-aware EMPCA and isotropic PCA under pink noise, timing-jitter family ($N_{\text{train}} = 400$, $N_{\text{test}} = 200$, mean $\pm$ std, 8 seeds).

Rank k	$\chi^2_{\text{EMPCA}}(k)$	$\chi^2_{\text{iso}}(k)$	EMPCA $\Delta\chi^2/\chi^2(1)$
1	1029.2 ± 0.9	1029.5 ± 0.9	–
2	1028.9 ± 0.9	1031.5 ± 1.0	-0.03%
3	1028.6 ± 0.9	1033.9 ± 0.9	-0.06%

F.7 Convergence behavior

Figures 25 and 26 track the normalized objective $\mathcal{L}^{(r)}/\mathcal{L}^{(0)}$ and the subspace principal angle over EM iterations, for ranks $k = 1, 2, 3$ under random and SVD warm-start initialization. Monotone non-increase holds for all seeds and ranks, consistent with Theorem C.1; SVD initialization converges in under 10 iterations for all ranks, versus up to 30–50 for random initialization at $k = 3$ (3–5$\times$ reduction), and the estimated convergence factors are consistent with the spectral gap of $\Sigma^{-1}\hat{R}_x$ (Appendix C.5).

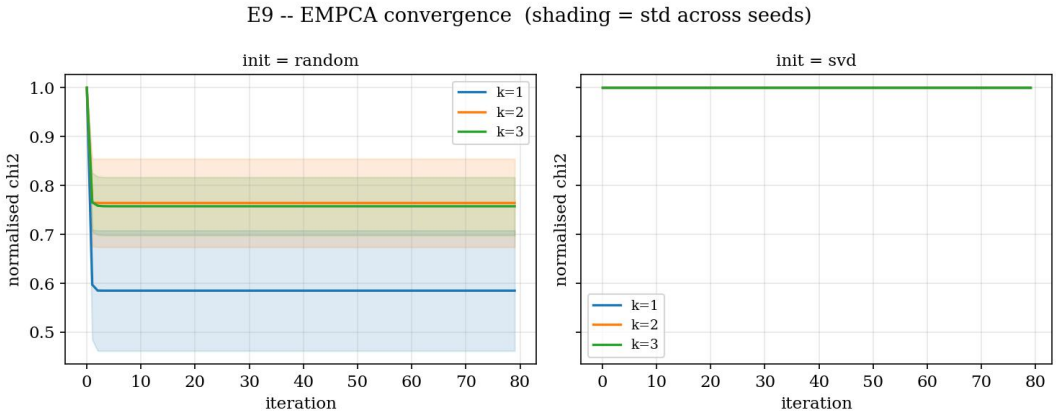


Figure 25: EMPCA convergence: normalized objective versus EM iteration for $k = 1, 2, 3$ under random initialization (left) and SVD warm-start (right); shaded bands = std across 8 seeds.

F.8 Noise-budget study (Study B)

Using the total measured PSD, the baseline sensitivity proxy is

$$I \propto \int df \, \frac{|S(f)|^2}{P_{\text{tot}}(f)}, \qquad \sigma_E \propto I^{-1/2}, \tag{72}$$

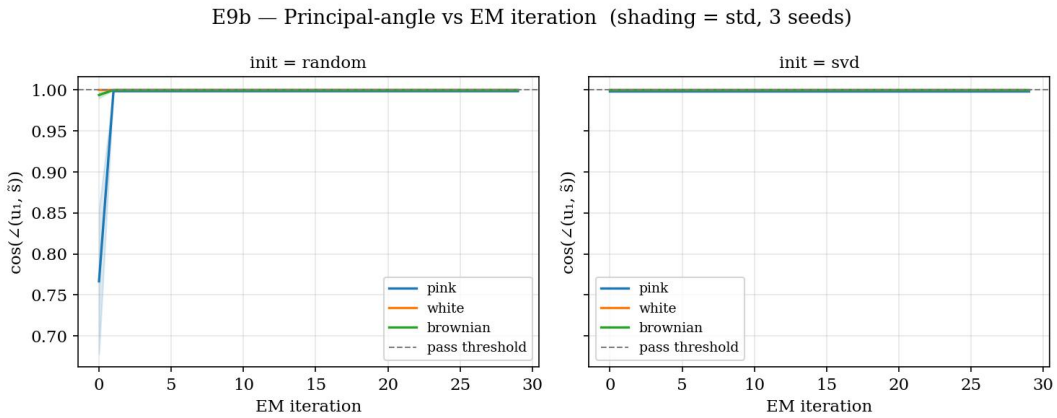


Figure 26: Principal-angle convergence over EM iterations for the same ranks and initializations as Figure 25.

up to the calibration convention of $S(f)$ (cf. Eq. (15)). The weighting density $W(f) = |S(f)|^2/P_{\mathrm{tot}}(f)$ (Figure 27) determines which bands dominate the information budget. Sensitivity is recomputed after removing or scaling individual noise components (Table 13), ranking engineering priorities for noise reduction.

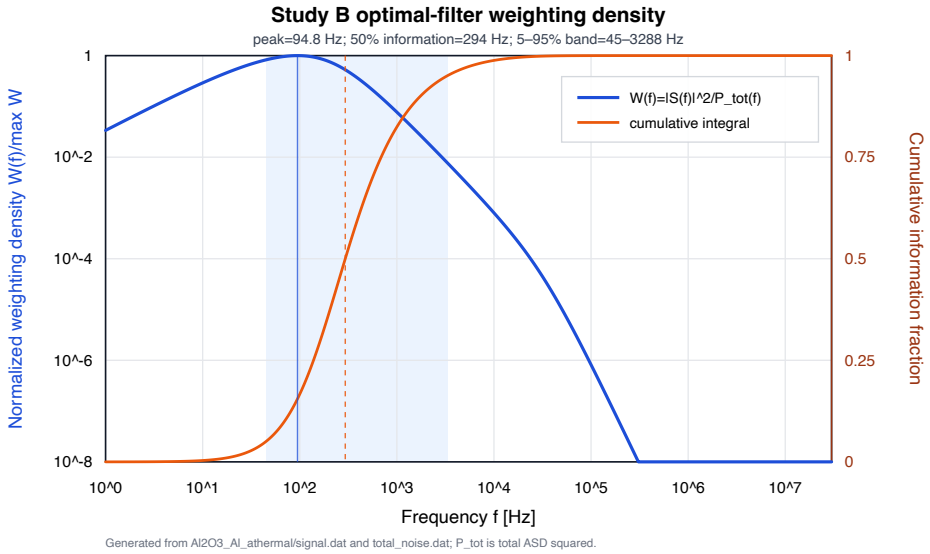


Figure 27: Optimal-filter weighting density $W(f) = |S(f)|^2/P_{\mathrm{tot}}(f)$ used in the noise-budget study.

Table 13: Predicted relative sensitivity for leave-one-out and scale-down noise scenarios.

Scenario	Relative sensitivity	Improvement vs baseline
Baseline (all components)	1.000	–
Remove pink ($1/f$) noise	1.031	+3.1%
Remove Brownian ($1/f^2$) noise	1.018	+1.8%
Remove white noise	1.004	+0.4%
Scale pink noise by 0.5×	1.014	+1.4%
Scale pink noise by 0.2×	1.025	+2.5%
Scale Brownian noise by 0.5×	1.009	+0.9%

G Detector Simulation and Preprocessing Details

G.1 LAMCAL and cryogenic detector response

The LAMCAL (LAboratory for Microcalorimeter CALibration) detector used in this study is a cryogenic bolometer operated at a base temperature of $T_{\mathrm{base}} \approx 20\,\mathrm{mK}$, below the superconducting transition of the sensor element. The detector consists of an absorber thermally coupled to a transition-edge sensor (TES) or metallic magnetic calorimeter (MMC) readout. Energy deposited in the absorber produces a temperature rise

$$\Delta T = \frac{\Delta E}{C(T)}, \tag{73}$$

where $C(T)$ is the heat capacity of the absorber–sensor system; the resulting change in sensor resistance (TES) or magnetization (MMC) is transduced into a voltage or flux signal by the readout chain.

The detector is read out through a SQUID amplifier chain whose noise enters the budget through $P_{\mathrm{SQUID}}(f)$; intrinsic sources include Johnson noise (P_{Johnson}), thermodynamic fluctuation noise (P_{TD}), and athermal excess noise (P_{Er}), combined in quadrature:

$$P_{\mathrm{tot}}(f) = P_{\mathrm{SQUID}}(f) + P_{\mathrm{Johnson}}(f) + P_{\mathrm{TD}}(f) + P_{\mathrm{Er}}(f). \tag{74}$$

The pulse template $s(t)$ is the mean response to a calibration-energy event, normalized to unit amplitude; it is the template s_0 used throughout the paper.

G.2 Quasiparticle signal formation

A particle depositing energy E first generates athermal phonons and quasiparticles on a timescale faster than the thermal response ($\lesssim 1\,\mu\mathrm{s}$); these propagate ballistically until absorbed by the sensor. Signal formation proceeds in three stages.

Energy partitioning. The deposited energy is distributed among phonon, triplet-state, UV-photon, and IR-photon channels by the material-dependent branching ratios of the `DELightSignalFormation` model,

$$E = E_{\mathrm{ph}} + E_{\mathrm{tr}} + E_{\mathrm{uv}} + E_{\mathrm{ir}}, \tag{75}$$

and each component is converted to discrete quanta by the corresponding mean excitation scale, e.g.

$$N_{\mathrm{uv}} = \lfloor E_{\mathrm{uv}}/E_{\mathrm{UV}} \rfloor, \qquad N_{\mathrm{ph}} \approx E_{\mathrm{ph}}/\epsilon_{\mathrm{ph}}, \qquad \epsilon_{\mathrm{ph}} \simeq 1\,\mathrm{meV}. \tag{76}$$

Collection and transport. Optical and triplet quanta are distributed across channels by a multinomial draw on position-dependent collection efficiencies. For phonons, the expected count in channel i is $\lambda_i(x,y,z) = \mathrm{PCE}_i(x,y,z)\, N_{\mathrm{ph}}$ from precomputed transport maps, with observed counts

$$N_{\mathrm{ph},i}^{\mathrm{obs}} \sim \mathrm{Poisson}(\lambda_i), \tag{77}$$

the dominant intrinsic-resolution term for phonon channels. Arrival times are sampled from the channel-specific distribution $P_{\mathrm{arr},i}(t \mid x,y,z)$ and histogrammed into digitizer bins of width $\Delta t = 1/f_s$:

$$H_i[n] = \sum_k \mathbf{1}\Big(t_{\mathrm{ph},i}^{(k)} \in [n\Delta t, (n{+}1)\Delta t)\Big). \tag{78}$$

Template shaping. Histogrammed counts are scaled by gain/calibration constants, $A_i[n] = G_i H_i[n]$, and convolved with the channel pulse template,

$$s_i[n] = (A_i * T_i)[n] = \sum_m A_i[m]\, T_i[n-m]. \tag{79}$$

Colored noise from the `NoiseGenerator` is then added, $x_i[n] = s_i[n] + \eta_i[n]$, optionally followed by digitization. The full chain

$$E_{\mathrm{ph}} \to N_{\mathrm{ph}} \to \lambda_i \to N_{\mathrm{ph},i}^{\mathrm{obs}} \to H_i[n] \to A_i[n] \to s_i[n] \to x_i[n] \tag{80}$$

ensures the simulated traces satisfy the generative model $x = as{+}n$, $n \sim \mathcal{N}(0,\Sigma)$ assumed throughout.

Simulator architecture. The `TraceSimulator` package separates detector response maps, energy partitioning, geometry handling, and noise generation into independent modules. A standard simulator produces baseline waveforms; an extended long-trace version additionally provides clean/noisy pairs and signal masks for analysis studies.

G.3 Noise generation

The `NoiseGenerator` produces time-domain traces consistent with a specified one-sided PSD $P(f)$. For trace length N and sampling frequency f_s, frequency bins are $f_k = kf_s/N$ and the discrete noise amplitudes are

$$\tilde{\eta}_k = \sqrt{\frac{N \cdot P(f_k)}{2\,\Delta f}}\, e^{i\phi_k}, \qquad \phi_k \sim \text{Uniform}[0, 2\pi), \tag{81}$$

with $\Delta f = f_s/N$; the factor $N/2$ implements Parseval normalization for the real inverse FFT, and the time-domain trace is $\eta[n] = \text{IRFFT}(\tilde{\eta}_k)$. Built-in spectral shapes are white, pink ($1/f$), Brownian ($1/f^2$), blue, and violet; a custom mode interpolates a user-supplied tabular PSD onto the frequency grid, used here to inject the measured detector spectra (SQUID, Johnson, TD, Er) into simulated traces. The resulting covariance is Toeplitz for stationary noise and diagonal in the frequency representation with $J_k = P(f_k)/(2\Delta f)$, recovering the standard OF weighting. Figure 28 audits the generator against its target PSD.

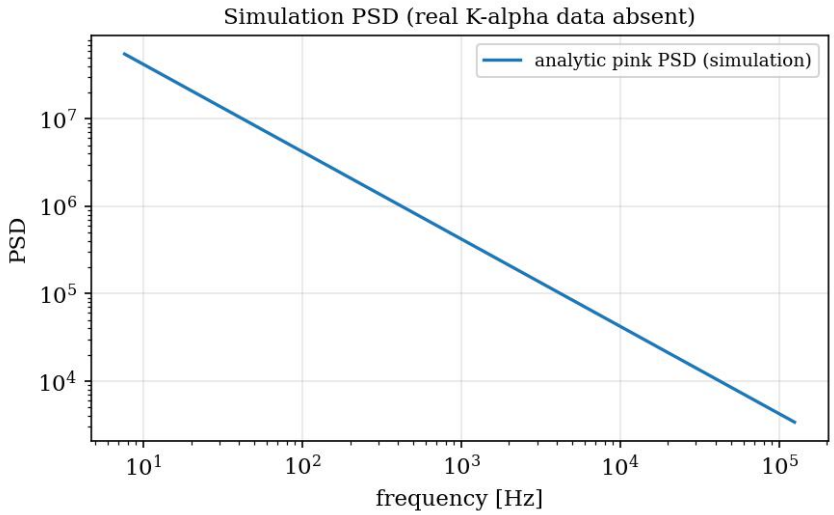


Figure 28: PSD audit: generated time-domain noise reproduces the target spectral density under the one-sided PSD convention used by the OF and EMPCA experiments.

G.4 Input spectra and working conventions

The noise characterization uses the total, SQUID, Johnson, TD, and Er component spectra, with an auxiliary MMC reference spectrum. The source figures label the vertical axis as $\sqrt{S_\phi}$ in $\mu\Phi_0/\sqrt{\text{Hz}}$, i.e. amplitude spectral densities (ASD); following the internal

convention $1\,\mathrm{V}/\Phi_0$, PSDs are constructed as $P_i(f) = N_i^2(f)$ where $N_i(f)$ is the ASD of component i. Before physics interpretation we verify frequency-grid consistency, monotonicity and finiteness of the spectra, and quadrature consistency of the total.

G.5 Study A: noise-budget decomposition

Study A determines which noise source dominates in each frequency band, where crossovers occur, and whether the detector is readout- or device-limited.

Quadrature consistency. The total ASD is tested against the quadrature sum of components,

$$N^2_{\mathrm{model}}(f) = N^2_{\mathrm{Er}}(f) + N^2_{\mathrm{Johnson}}(f) + N^2_{\mathrm{SQUID}}(f) + N^2_{\mathrm{TD}}(f), \tag{82}$$

by comparing $N_{\mathrm{model}}(f)$ to the provided total on the native frequency grid (Figure 29).

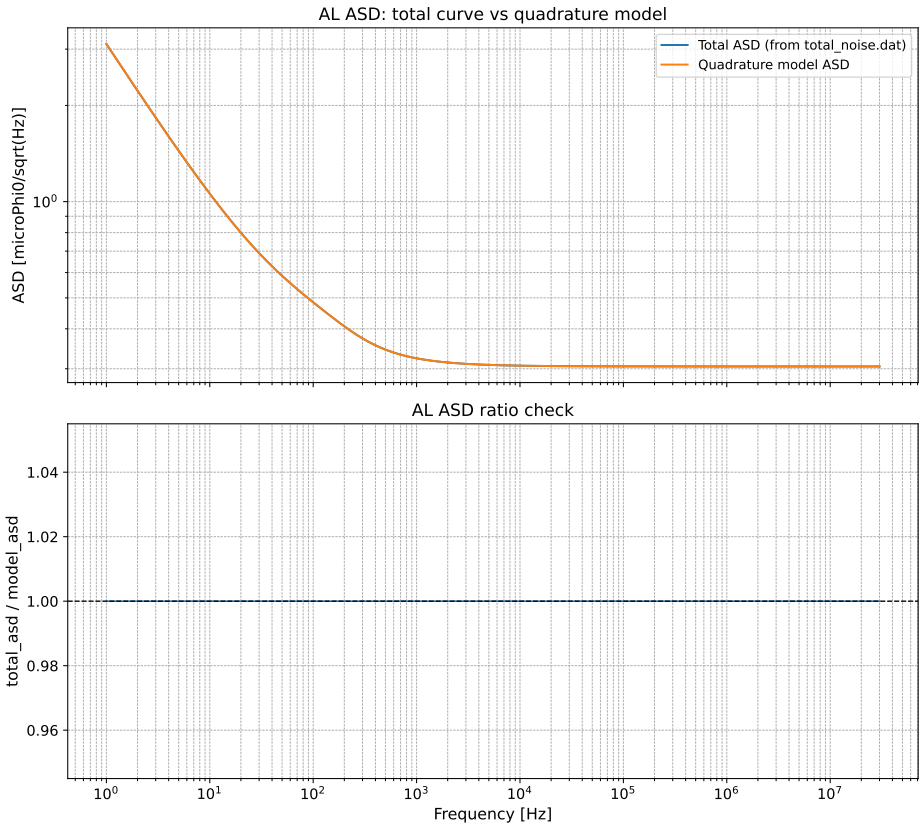


Figure 29: Study A quadrature validation: reconstructed total ASD from the component quadrature sum versus the provided total ASD.

Component budget and fractional contributions. Figure 30 shows the total and component ASDs; Figure 31 shows the PSD-based fractional contributions $R_i(f) = P_i(f)/P_{\mathrm{tot}}(f)$, with $R_i \approx 1$ indicating dominance of component i.

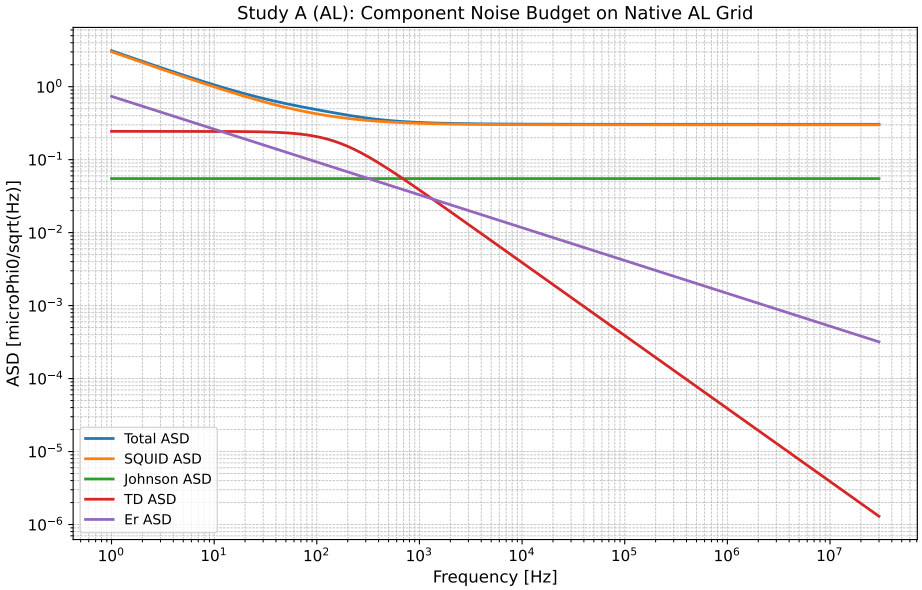


Figure 30: Study A overview of the noise budget on the native frequency grid.

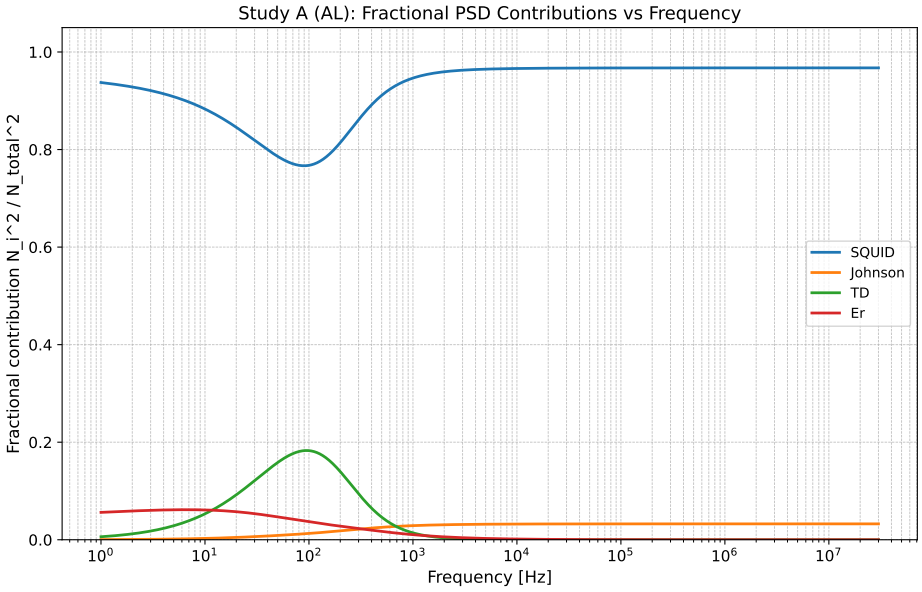


Figure 31: Fractional PSD contributions of the noise components versus frequency.

Dominance bands and crossovers. Tables 14 and 15 summarize the dominant source per logarithmic band (median fractional contribution) and the crossover frequencies solving $P_i(f) = P_j(f)$ for selected pairs.

Table 14: Dominant noise source by frequency band (Study A).

Band	Dominant	Secondary	Dominant fraction	Interpretation
$< 3\,\mathrm{Hz}$	$1/f$ phonon	Vibrational	78%	Thermal fluctuation dominated
3–30 Hz	$1/f$ phonon	TES Johnson	60%	Low-frequency crossover regime
30–300 Hz	TES Johnson	SQUID readout	58%	Electronic noise crossover
300–3000 Hz	SQUID readout	TES Johnson	72%	Readout-limited band
$> 3000\,\mathrm{Hz}$	SQUID readout	—	91%	High-frequency readout floor

Table 15: Crossover frequencies between selected noise components.

Component pair	Crossover frequency [Hz]	Comment
$1/f$ phonon vs. TES Johnson	≈ 18	Below: phonon-dominated; above: electronic
TES Johnson vs. SQUID readout	≈ 240	Below: Johnson-dominated; above: SQUID
$1/f$ phonon vs. SQUID readout	≈ 8	Combined crossover reference point
Vibrational vs. $1/f$ phonon	≈ 0.5	Vibrational excess confined to sub-Hz

Status. Study A provides the validated noise-budget interpretation for the primary sensor chain. The provisional MMC comparison is kept separate because the physical normalization of the raw MMC spectrum has not been independently verified.

H List of Symbols

Notation used throughout the paper. Complex notation with † (conjugate transpose) is used by default; all statements specialize to real data with $\dagger \to T$. Vectors are columns; whitened quantities carry tildes.

Category	**Symbol**	**Meaning**
Dimensions	d	Trace length (samples or frequency bins per event).
	N	Number of events in the dataset.
	k	Rank of the learned signal subspace (also frequency-bin index; clear from context).

Category	Symbol	Meaning
	C	Number of detector channels.
Indices	i	Event index, $i = 1, \ldots, N$.
	k	Frequency-bin index of the DFT.
	c	Channel index, $c = 1, \ldots, C$.
	r	EM iteration index.
Data	$x,\ x_i$	Observed trace (column vector in $\mathbb{C}^d$).
	X_k	Frequency-domain coefficient of the observed trace at bin k.
	X	Data matrix, rows $x_i^\dagger$ ($N \times d$).
	$\tilde{x},\ \tilde{X}$	Whitened trace / data matrix, $\tilde{x} = \Sigma^{-1/2} x$.
Signal model	$s_0,\ S_k$	Fixed signal template (time / frequency domain).
	$\hat{s}$	Reconstructed signal.
	$A,\ a_i$	Signal amplitude (pulse height) of an event.
	t_0	Signal arrival time (time offset).
	U	Learned rank-k basis, $U \in \mathbb{C}^{d \times k}$, $U^\dagger \Sigma^{-1} U = I_k$.
	$c,\ C$	Latent coefficient vector / matrix for the rank-k model.
	P	Decoder basis of the noise-aware linear AE, $P^\dagger \Sigma^{-1} P = I_k$.
	z	Latent code of the autoencoder, $z = P^\dagger \Sigma^{-1} x$.
Noise	$n,\ N_k$	Additive noise (time / frequency domain).
	Σ	Noise covariance matrix ($\succ 0$); diagonal in frequency space for stationary noise.
	Σ^{-1}	Inverse covariance; the observation-space metric.
	$\Sigma^{-1/2}$	Whitening transform.
	$J(f),\ J_k$	Noise power spectral density (PSD).
Optimal filtering	χ^2	Mahalanobis / weighted least-squares objective, Eq. (4).
	$\Phi_k,\ \Phi_k^{(0)}$	Unnormalized / normalized OF kernels.

Category	Symbol	Meaning
	N_Φ	Filter normalization $\sum_k \lvert S_k\rvert^2/J_k$; total SNR budget.
	$\sigma_A,\ \sigma_E$	Amplitude / energy resolution.
	$B(t_0)$	Delayed matched-filter response (time-shift OF).
Subspace estimation	$\hat{R}_x$	Sample covariance $N^{-1}\sum_i x_i x_i^\dagger$.
	λ_j	Eigenvalues of $\Sigma^{-1}\hat{R}_x$ (descending).
	Δ_λ	Spectral gap between signal and noise eigenvalues.
	$\rho_w,\ \theta_w$	Weighted (whitened-space) subspace cosine / angle.
	$P_{\tilde{U}}$	Orthogonal projector onto the whitened subspace.
	$\mathrm{Gr}(k,d)$	Grassmannian of k-dimensional subspaces.
Symmetry	G_{noise}	Noise isometry group preserving Σ^{-1} (Definition E.1).
	$\mathcal{U}(k)$	Latent gauge group of basis rotations (Definition E.2).